

AKTU



B. Tech II-Year

Gateway Classes



Semester -IV

CS IT & Allied Branches

BCS402 Theory of Automata and Formal Languages

UNIT-2 Regular Expressions and Languages



Gateway Series for Engineering

- ✓ Topic Wise Entire Syllabus
- ✓ Long - Short Questions Covered
- ✓ AKTU PYQs Covered
- ✓ DPP
- ✓ Result Oriented Content



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Gateway Classes



BCS402 Theory of Automata and Formal Languages

Unit-2

Introduction to Regular Expressions and Languages

Syllabus

Regular Expressions and Languages: Regular Expressions, Transition Graph, Kleen's Theorem, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem, Regular and Non-Regular Languages- Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability- Decision properties, Finite Automata and Regular Languages



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- A regular language is a language that can be expressed with a regular expression or a deterministic or non-deterministic finite automata or state machine.

Whether the below following language are regular or not

- $L = \{a^m b^n \mid m, n \geq 0\}$ yes (able to learn order)
x No comparison, infinite
- $L = \{a^p b^q c^r \mid p, q, r \geq 0\}$ yes *Order, No comparison, infinite*
- $L = \{a^m b^n \mid 1 \leq m \leq 500, 1 \leq n \leq 1000\}$ yes {language is finite, able to learn order}
- $L = \{a^n b^n \mid n \geq 0\}$ no {able to do counting / comparison, infinite}
- $L = \{a^n b^n \mid 1 \leq n \leq 10\}$ yes finite, *Comp*
- $L = \{a^n b^n \mid 1 \leq n \leq 2^{29} \text{ prime number}\}$ yes {comparison, finite}
- $L = \{a^n b^n \mid 1 \leq n \leq 2^{|gate|}\}$ yes *Counting*

➤ A way of representing regular language

➤ Expression of string and operator

➤ *Kleene closure $[a^*]$

$\{\epsilon, a, aa, aaa, \dots\} \rightarrow \{a^0, a^1, a^2, a^3, a^4, \dots\}$
 $\{a, aa, aaa, aaaa, \dots\} \rightarrow \{a^1, a^2, a^3, a^4, \dots\}$

➤ + positive closure $[a^+]$

➤ . Concatenation $[a.b]$

cde fg
h
 cde.fg

➤ + union $[a+b]$

a+b
 $\{a, b\}$

➤ $(a+b).(a+b)$

$(a+b).(a+b) = (a+b)^2$ - double length all string

➤ $(a+b)^* = \{(a+b)^0, (a+b)^1, (a+b)^2, (a+b)^3, \dots\}$

aa
 ab
 ba
 bb

aa
 ab
 ba
 bb

$\{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, aba, \dots\}$

$$a \cdot \epsilon = a$$

$$1. r = a, L(r) = \{a\}$$

$$2. r = a+b, L(r) = \{a, b\}$$

$$3. r = a.b, L(r) = \{ab\}$$

$$4. r = a+b+c, L(r) = \{a, b, c\}$$

$$5. r = \{ab+a\}.b, L(r) = \{abbb, abb\}$$

$$6. r = a^+, L(r) = \{a, aa, aaa, \dots\}$$

$$7. r = a^*, L(r) = \{\epsilon, a, aa, aaa, \dots\}$$

$$8. r = (a+ba).(b+a), L(r) = \{ab, aa, bab, baa\}$$

$$9. r = \Phi, L(r) = \{\}$$

$$10. r = \epsilon, L(r) = \{\epsilon\}$$

$$11. (a+b).bb, \{abbb, bbbb\}$$

$$12. r = (a+b)^*, L(r) = \{\epsilon, a, b, ab, ba, bb, \dots\}$$

$$13. r = (a+b)^*(a+b), L(r) = (a+b)^+ = \{a, b, aa, ba, \dots\}$$

$$14. r = \underline{a^*}.a^*, L(r) = a^* = \{\epsilon, a, aa, aaa, \dots\}$$

$$15. r = (ab)^*, L(r) = \{\epsilon, ab, abab, ababab, \dots\}$$

$$16. r = (a+b)^2, L(r) = \{aa, ab, ba, bb\}$$

$$(a+b)^3$$

$$(a+b)(a+b)(a+b)$$

$$\{aa, ab, ba, bb\}(a+b)$$

$$\{aaa, aba, baa, bba,$$

$$aab, abb, bab, bbb\}$$

Find out the regular expression for the following regular language

 $\Sigma = \{a, b\}$

1. $L = \{\text{start with ab}\} \rightarrow ab(a+b)^*$
2. $L = \{\text{start with bba}\} \rightarrow bba(a+b)^*$
3. $L = \{\text{end with abb}\} \rightarrow (a+b)^*abb$
4. $L = \{\text{contain a substring aab}\} \rightarrow (a+b)^*aab(a+b)^*$
5. $L = \{\text{start with and ends with a}\} \rightarrow a + a(a+b)^*a$
6. $L = \{\text{start and end with the same symbol}\} \rightarrow a + b + a(a+b)^*a + b(a+b)^*b$
7. $L = \{\text{start with and ends with the different symbol}\} \rightarrow a(a+b)^*b + b(a+b)^*a$
8. $L = \{|w| = 3\} \rightarrow (a+b)(a+b)(a+b)$

9. $L = \{|w| \geq 3\} \rightarrow (a+b)(a+b)(a+b)(a+b)^*$

10. $L = \{|w| \leq 3\} \rightarrow \epsilon + (a+b) + (a+b)^2 + (a+b)^3$

Find out the regular expression for the following regular language

 $\Sigma = \{a, b\}$

1. $L = \{ |w|_a = 2 \}$ $b^* a b^* a b^*$
2. $L = \{ |w|_a \geq 2 \}$ $(a+b)^* a (a+b)^* a (a+b)^*$
3. $L = \{ |w|_a \leq 2 \}$ $b^* + b^* a b^* + b^* a b^* a b^* / b^* (a+\epsilon) b^* (a+\epsilon) b^*$
4. 3rd symbol from the left end is b $(a+b)^2 b (a+b)^*$
5. 2nd symbol from the right end is a $(a+b)^* a (a+b)^{2+}$
6. $|w| \equiv 0 \pmod 3$ $[(a+b)^3]^*$
7. $|w| \equiv 2 \pmod 3$ $(a+b)^2 [(a+b)^3]^*$
8. $|w|_b \equiv 0 \pmod 2$ $a^* (a^* b a^* b a^*)^* / a^* + (a^* b a^* b a^*)^*$

9. $|w|_a \equiv 1 \pmod 3$ 10. $|w|_b \equiv 2 \pmod 3$

$b^* a b^* (b a b^* a b^* a b^*)^*$

$a^* b a^* b a^* (a^* b a^* b a^* b a^*)^*$

$\Sigma = \{a, b\}$

1. Language accepting the string of length 2

$$(a+b) \cdot (a+b) \mid (a+b)^2$$

2. Language accepting the string of length at least 2

$$(a+b) \cdot (a+b) \cdot (a+b)^*$$

3. Language accepting the string of length at most 2

$$\rightarrow \epsilon + a + b + (a+b) \cdot (a+b)$$

$$\rightarrow \epsilon + a + b + (a+b)^2$$

$$\rightarrow (\epsilon + a + b) \cdot (\epsilon + a + b)$$

$\Sigma = \{0, 1\}$

1. That starts either with 01 or end with 01

$$01(0+1)^* + (0+1)^*01$$

2. having at least two zero

$$(0+1)^* 0 (0+1)^* 0 (0+1)^*$$

two consecutive zero $(0+1)^* 00 (0+1)^*$

3. Having two consecutive zero or one

$$(0+1)^* 00 (0+1)^* \mid (0+1)^* 11 (0+1)^*$$

Φ, ϵ are primitive regular expression

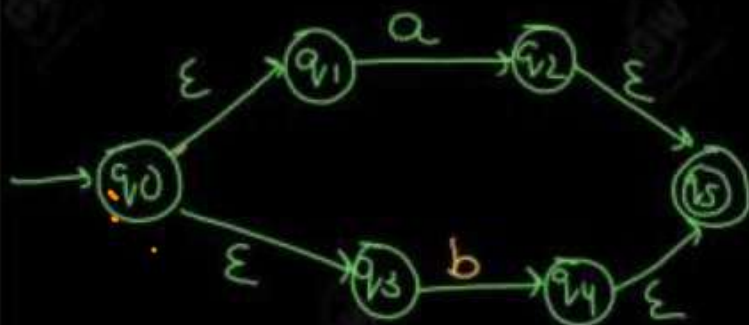
Φ



ϵ



$a+b$

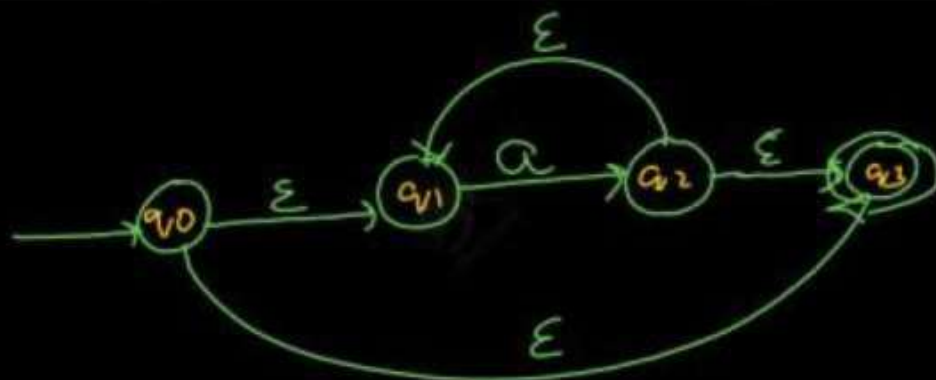


$a.b$

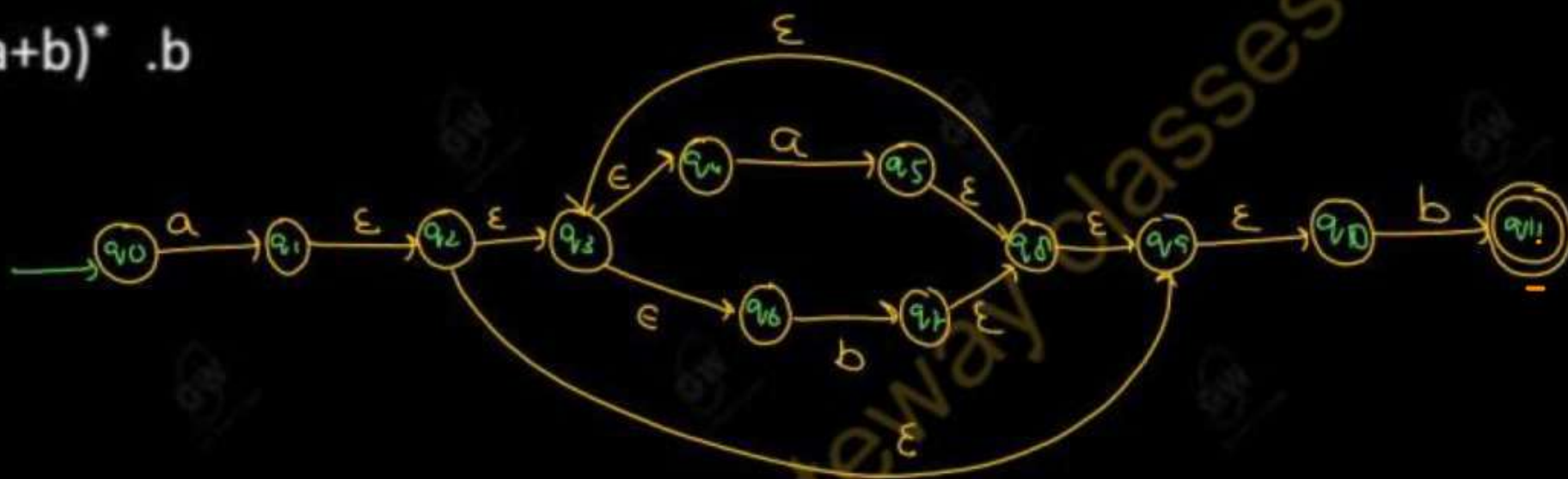


Convert regular expression to epsilon NFA

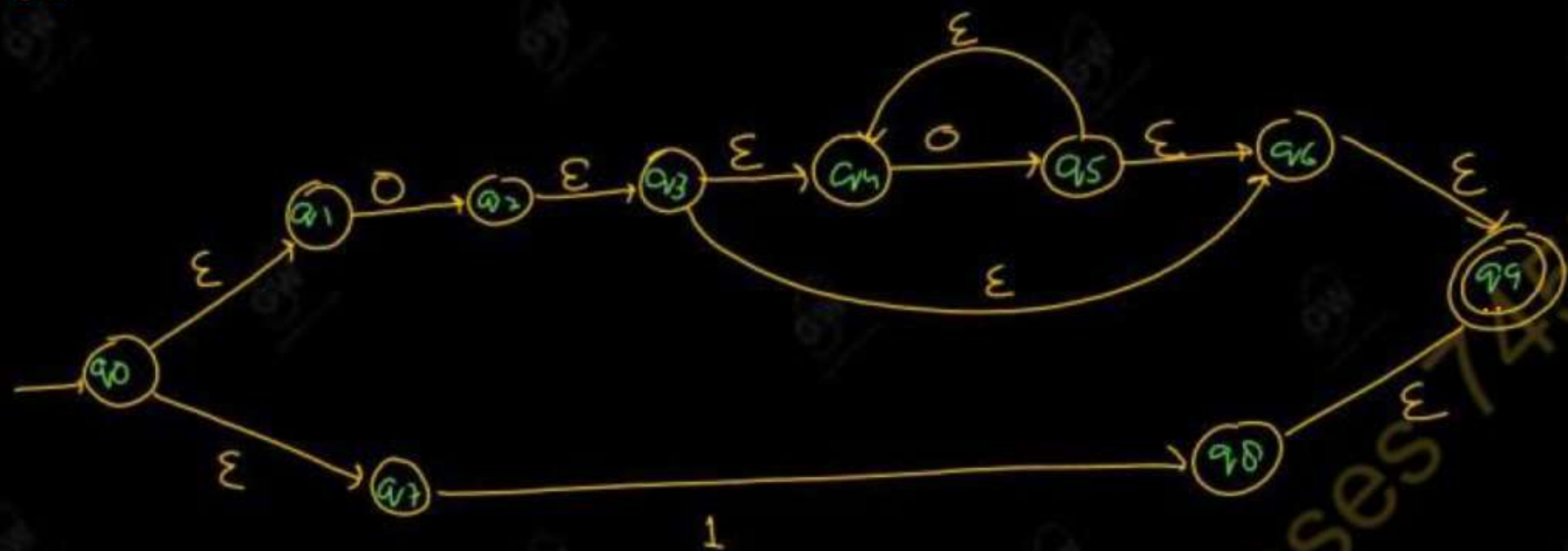
a^*



$a.(a+b)^*.b$



Convert regular expression to epsilon NFA

$$\frac{00^* + 1}{a + b}$$


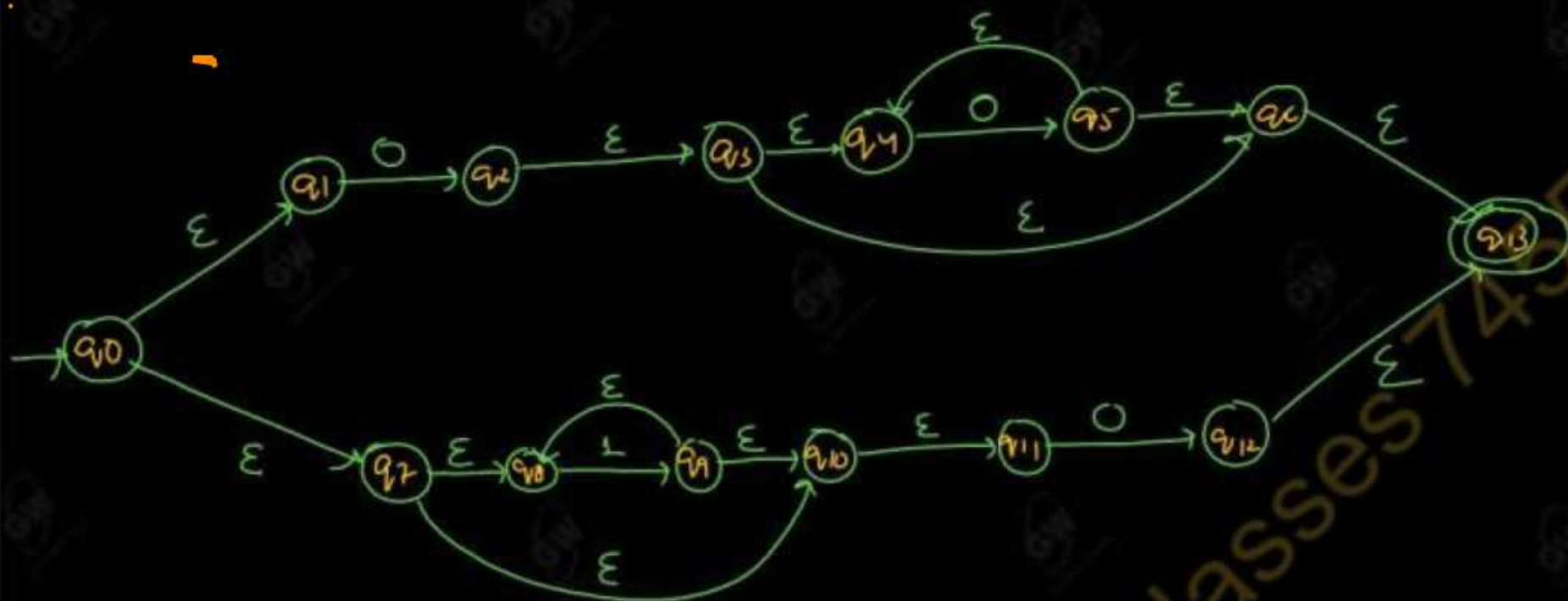
Convert the following Regular Expression into DFA (Finite Automata)

a^*

- epsilon NFA
- epsilon NFA to NFA
- NFA to DFA
- Minimization of DFA

- ε NFA
- ε-NFA to DFA
- Minimization of DFA

Convert regular expression to epsilon NFA

 $00^* + 1^*0$ 

$(0+1)^* 1 (0+1)$



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Convert regular expression to epsilon NFA

$(0+1^*)2^*3^*$



$(a|b)^*abb$ or $(a+b)^*abb$

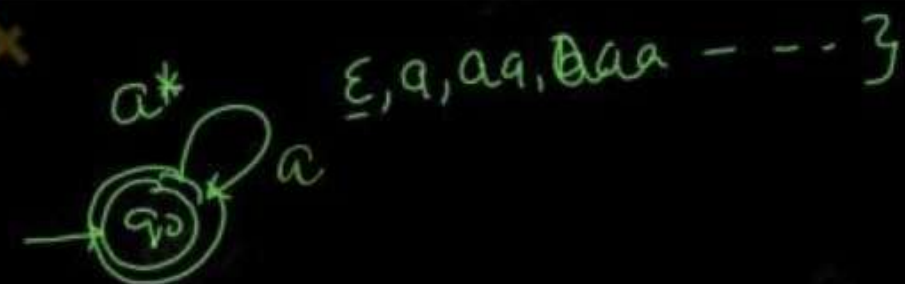


Convert regular expression to NFA

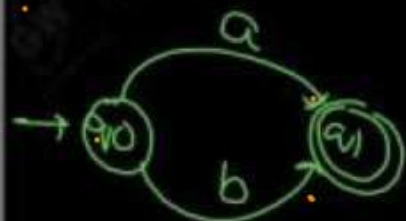
a



ϵ



a+b

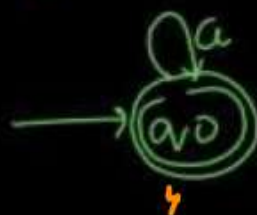


a.b



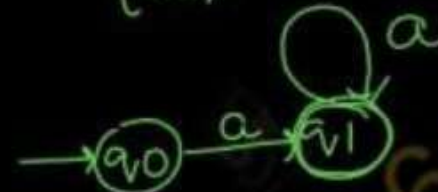
Convert regular expression to NFA

a^* $\{\epsilon, a, aa\}$

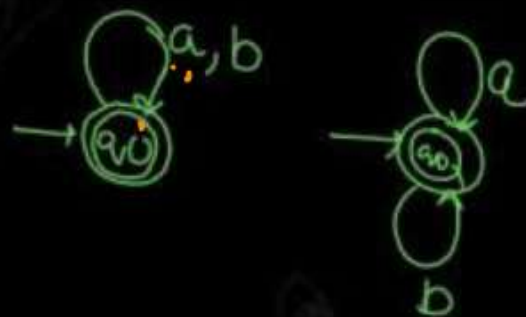


a^+

$\{a, aa, aaa, \dots\}$



$(a+b)^*$

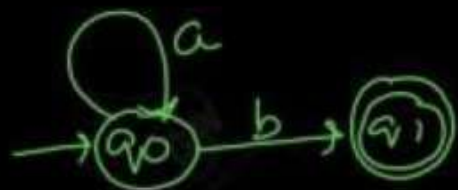


$(a.b)^*$



Convert regular expression to NFA

a^*b



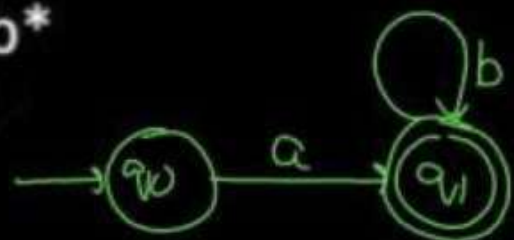
①

$a^*b(a+b)^*$



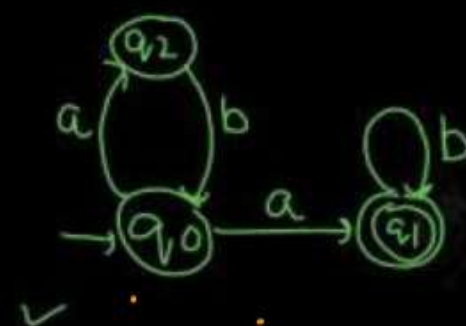
②

ab^*



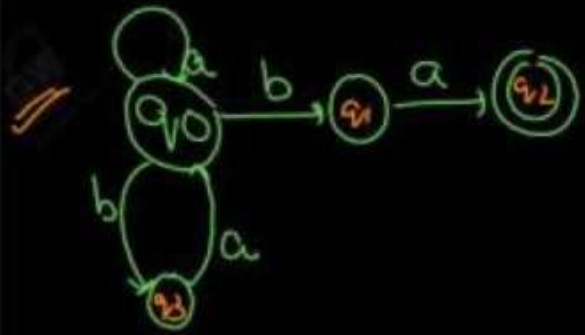
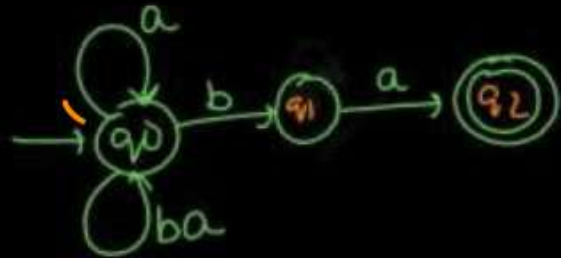
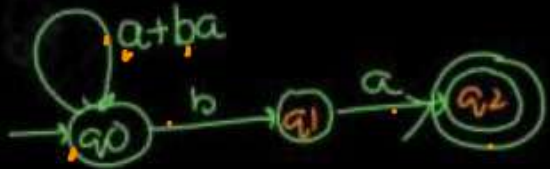
$(ab)^*$	a	b^*
ϵ	ϵ	$\epsilon \rightarrow a$
ab	b	
$abab$	bbb	
$ababab$	$bbbb$	

$(a.b)^*ab^*$

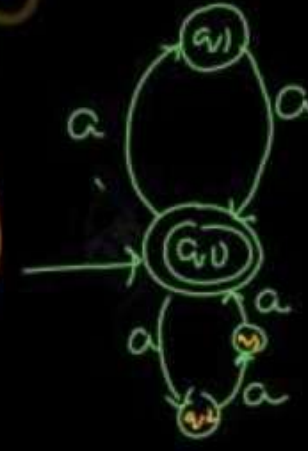
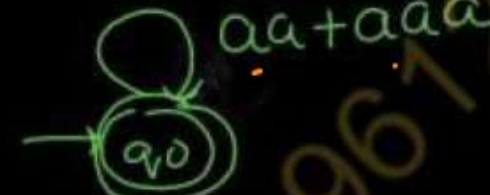


Convert regular expression to NFA

$(a+ba)^*ba$

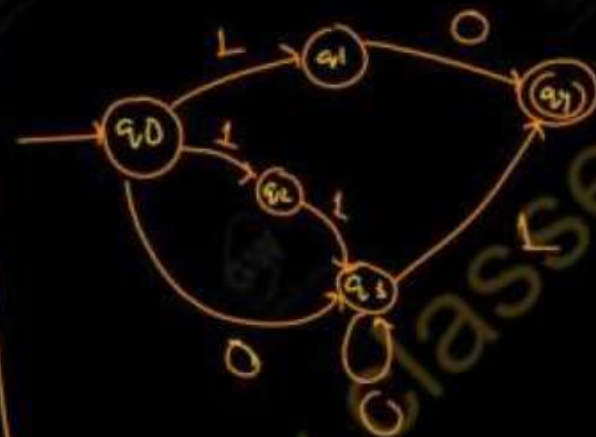
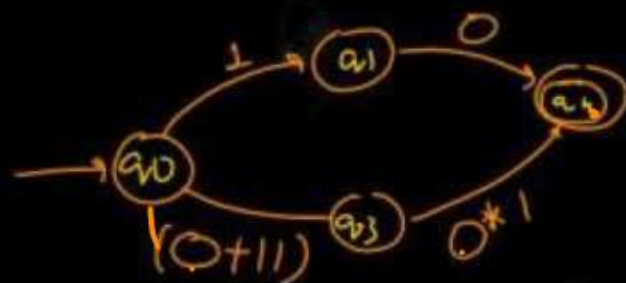
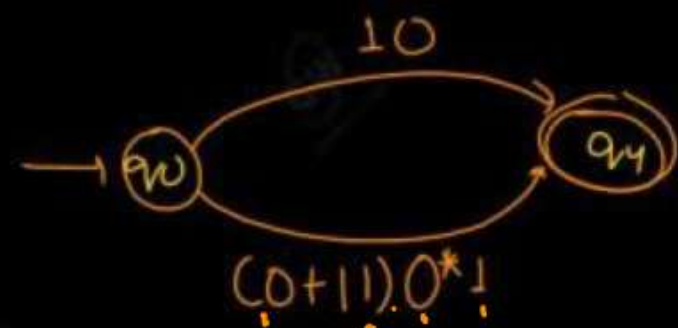


$(aa+aaa)^*$



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Convert regular expression to NFA

 $10+(0+11)0^*1$ 

Convert regular expression to NFA

 $1(1^*01^*01^*)^*$ 

Convert regular expression to NFA

 $1(1+10)^*+10(0+01)^*$ 

CLOSURE PROPERTIES OF REGULAR LANGUAGE

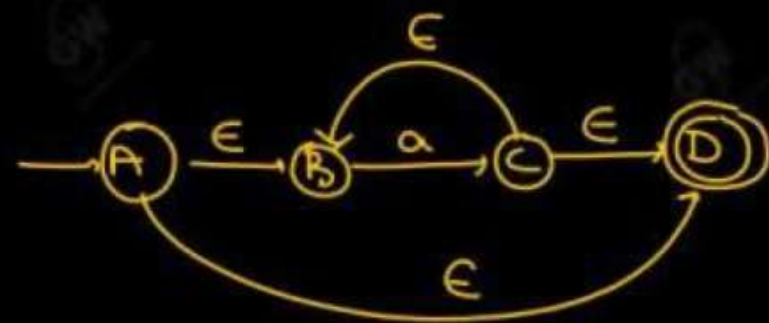
1 The union of two regular language is regular

$\Sigma = \{a, b\}$ $\Sigma^* = \{\epsilon, a, b, ab, ba, aa, \dots\}$

$L_1 = \{\epsilon, a, aa, aaa, aaaa, \dots\}$

R.E $\rightarrow a^* (L_1)$

ϵ -NFA

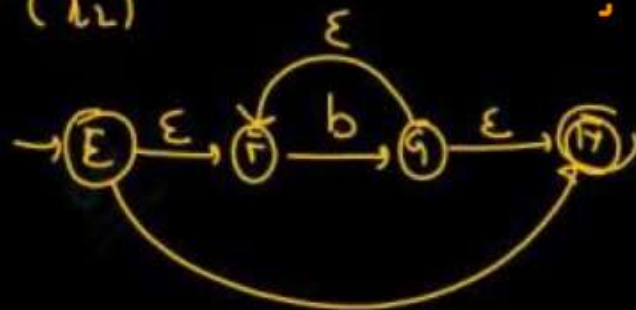


L_1 is regular language b/c we can make ϵ -NFA R.E

$L_2 = \{\epsilon, b, bb, bbb, bbbb, \dots\}$

R.E = $b^* (L_2)$

ϵ -NFA



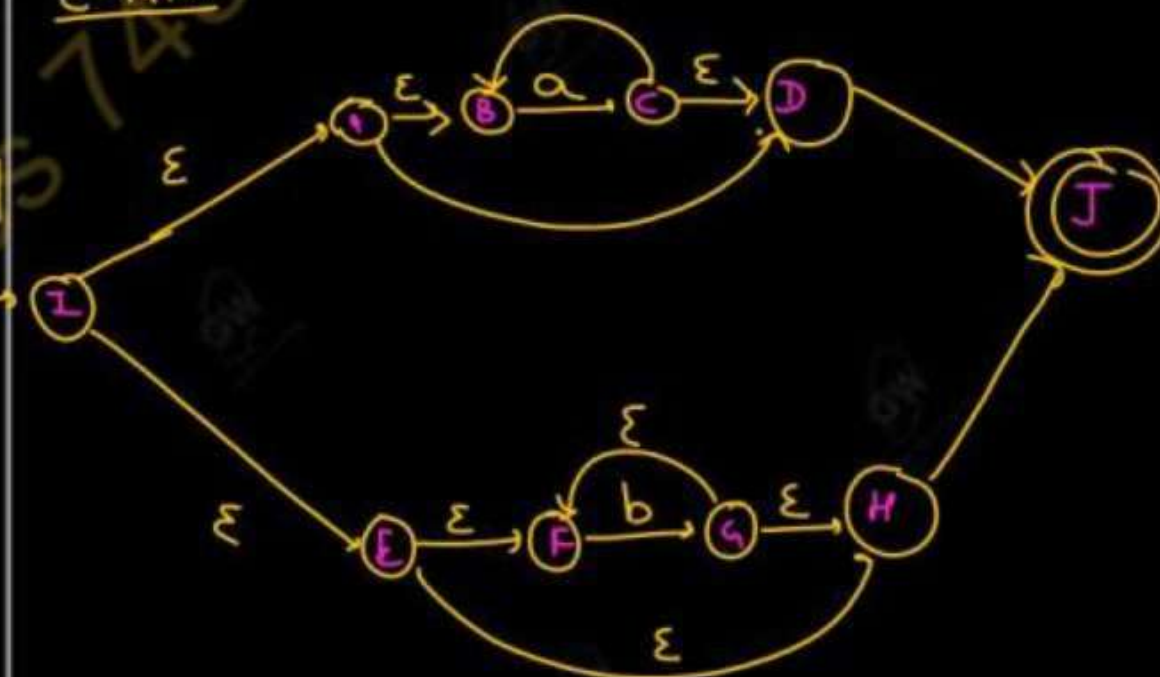
L_2 is regular language

$L_1 \cup L_2 = \text{Regular}$

$L_1 \cup L_2 = \{a^* + b^* (L_3)\}$

$L_1 \cup L_2 = \{\epsilon, a, b, aa, bb, aaa, bbb, \dots\}$

ϵ -NFA



$\rightarrow$ union of two Regular Lang is Regular

$L_3 = L_1 + L_2$
 $\uparrow$
Regular

2. The complement of the regular language is also regular

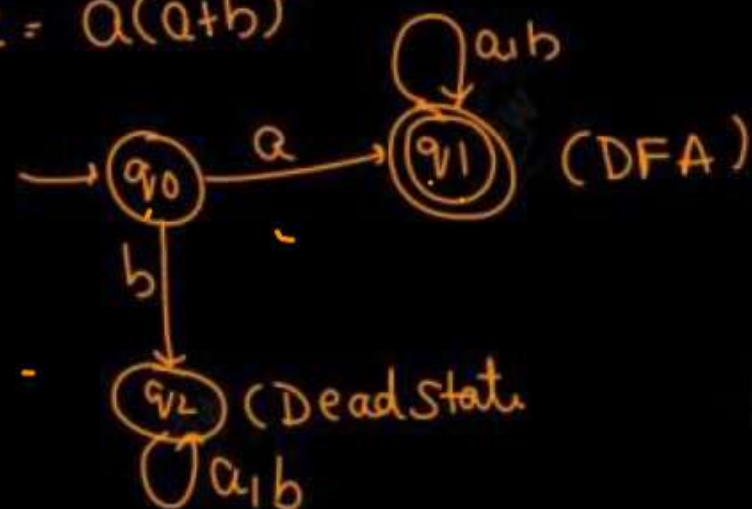
$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, \dots\}$$

$$L_1 = \{a, ab, aab, abaa, \dots\}$$

$$L_1 = \{w \mid w \text{ starts with } a\}$$

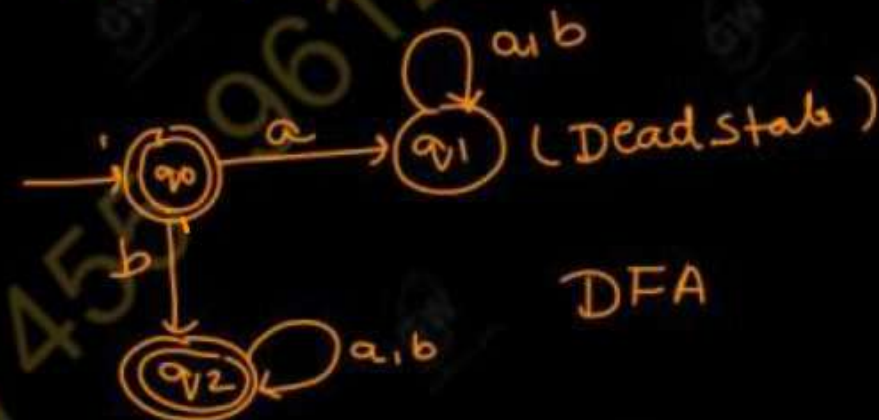
$$R.E = a(a+b)^*$$



L_1 is regular language

$$\overline{L_1} = \{\epsilon, b, ba, bb, baba, \dots\}$$

$\overline{L_1} = \{w \mid w \text{ is not start with } a\}$



$$R.E \quad \epsilon + b(a+b)^*$$

Complement of Regular language is Regular.

3. The intersection of two regular language is also regular

$$L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$$

$$A \cdot B = \overline{\overline{A} + \overline{B}}$$

L_1, L_2 is a Regular language
 $\overline{L_1}, \overline{L_2}$ are complement

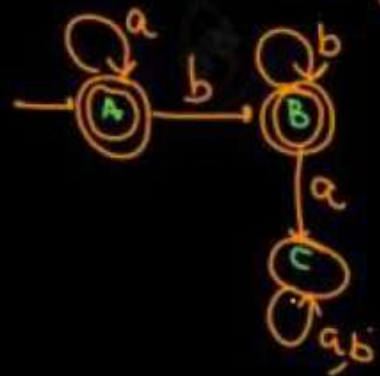
Reg $\rightarrow$ Regular

$$\overline{\text{Reg}} \cup \overline{\text{Reg}}$$

$$\overline{\text{Reg} \cup \text{Reg}} = \overline{\text{Reg}} = \text{Reg}$$

$$L_1 = \{a^n b^m, n \geq 0, m \geq 0\}$$

$$L = \{a, aa, \dots, ab, abb, \dots, b, bb, bbb, \dots\}$$



$$R.E = a^* b^*$$

L_1 is Regular

$$L_2 = \{a^n | n \geq 0\}$$

$$L_2 = \{\epsilon, a, aa, aaa, \dots\}$$

L_2 is regular language



$$R.E = a^*$$

$$L_1 \cap L_2 = \{a^n | n \geq 0\}$$

$$R.E = a^*$$



The intersection of two Regular lang is Regular

4. The difference of two regular language is also regular

$$L_1 - L_2 = L_1 \cap \bar{L}_2$$

$L_1 \& L_2$ is Regular language

→ Regular $\cap$ Regular

∴ the complement of Regular lang is Regular

→ Regular $\cap$ Regular

→ Regular

$$\Sigma = \{a\}$$

$$L_1 = \{\epsilon, a, aa, aaa, aaaa, \dots\}$$



$$R.E = a^*$$

$L_1 = \text{Regular language}$

$$L_2 = \{aa, aaaa, aaaaaa, \dots\}$$

$$(aa)^+ \rightarrow R.E$$

$L_2 = \text{Regular language}$



$$L_1 - L_2 = \{\epsilon, a, aaa, aaaaa, \dots\}$$



$$\epsilon, a, aaaa, aaaaaa, aaaaaaaa$$

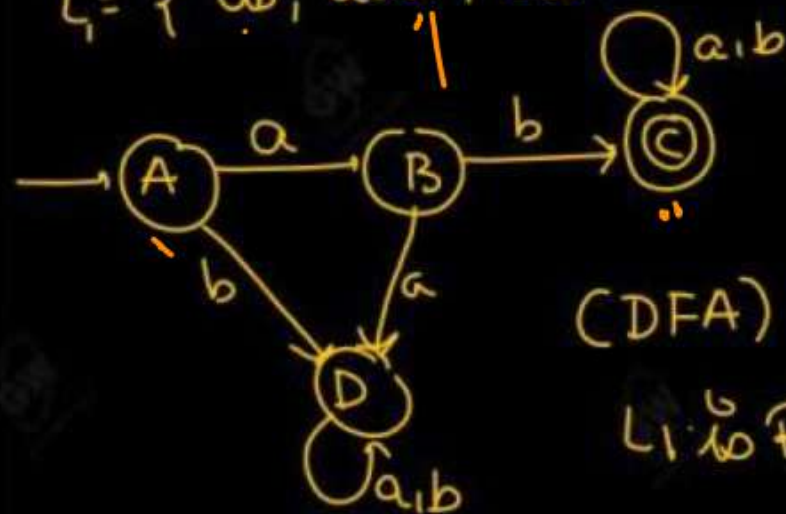
$L_1 - L_2 = \text{Regular lang}$

5. The reversal of a regular language is also regular

$$\Sigma = \{a, b\}$$

$$L = \{w \mid w \text{ starts with } a, b\}$$

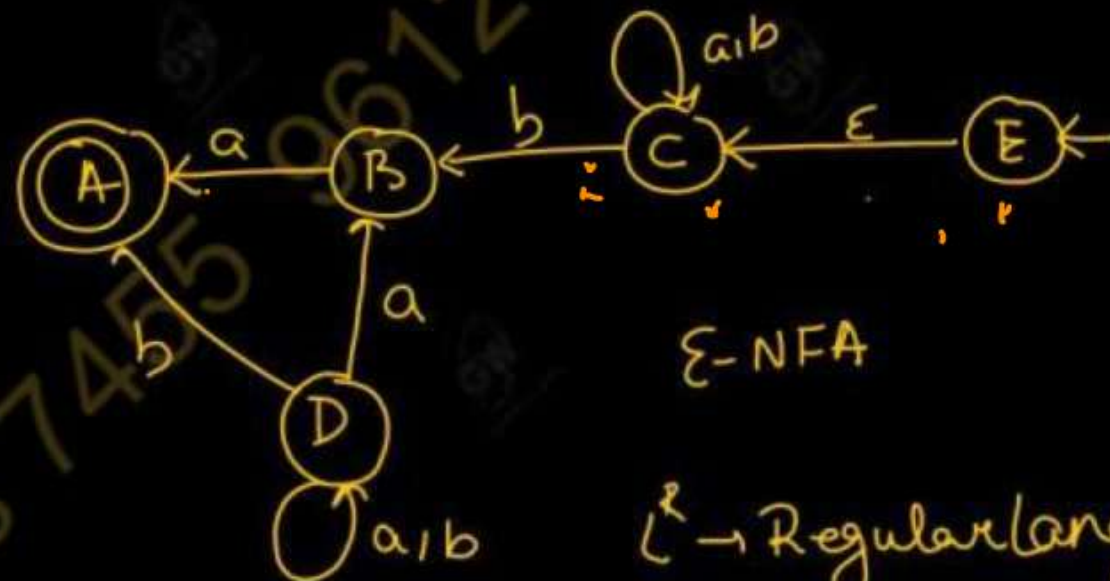
$$L = \{ab, aba, abaa, \dots, abbb, \dots\}$$



(DFA)

L is Regular

$$L^R = \{ba, aba, baa, \dots, abbb, \dots\}$$



ϵ -NFA

$L^R \rightarrow$ Regular lang

6. The Kleene closure of a regular language is also regular

$$L = \{a\}$$

$$r_1 = a$$

$$r_1^* = a^*$$

$$L = \{\epsilon, a, aa, \dots\}$$



7. The Concatenation of two regular language is also regular

$$L_1 = \{a\}$$

$$L_2 = \{b\}$$

$$r_1 \cdot r_2 = a \cdot b$$

$$L_3 = \{ab\}$$

L_3 is also regular language



8. A homomorphism (substitution of string for symbol) of a regular language is regular

$$h: \Sigma \rightarrow \Gamma^*$$

$$\Gamma = \{a, b\}$$

$$\Sigma = \{0, 1, 2\}$$

$$h(a) = 010$$

$$h(b) = 102$$

$$h(\varepsilon) = \varepsilon$$

$$h(xy) = h(x) \cdot h(y)$$

$$L_1 = \{a^n b^m \mid n, m \geq 0\}$$

$$a^* b^* - R.E$$

L is Regular



$$h(a) = x$$

$$h(b) = y$$

$$L_2 = \{x^n y^m \mid n, m \geq 0\}$$



L_2 is also Regular

8. The inverse homomorphism of a regular language is also regular

$$\Sigma = \{a, b\}$$

$$\Gamma^* = \{0, 1, 2\}$$

$$h(a) = 0$$

$$h(b) = 12$$

$$L = \{012120\}$$

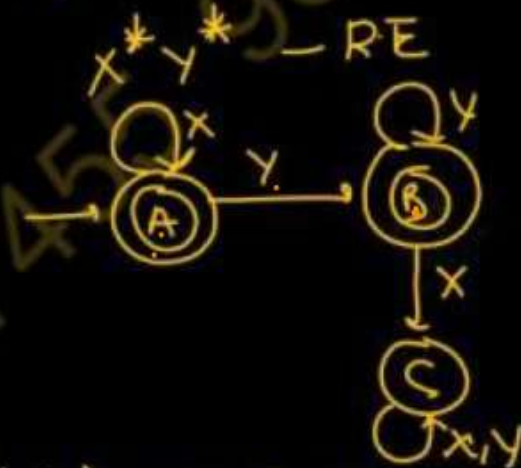
$$h^{-1}(L) = \{a, b, a\}$$

h^{-1} is inverse of homomorphism

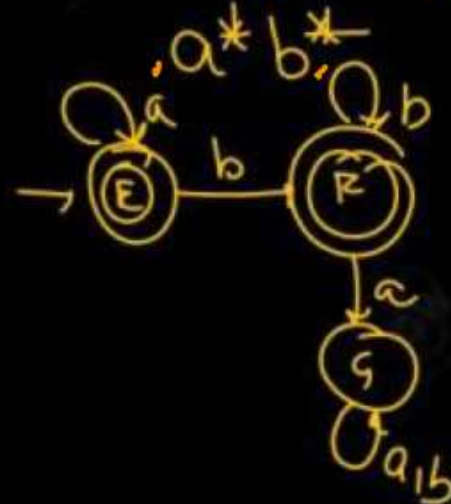
$$h(a) = x$$

$$h(b) = y$$

$$L = \{x^n y^m, n, m \geq 0\}$$



$$h^{-1}(L) = \{a^n b^m, n, m \geq 0\}$$



1. $\emptyset + r = r$

2. $\emptyset.r = r.\emptyset = \emptyset$

3. $\epsilon.r = r.\epsilon = r$

4. $\epsilon^* = \epsilon$ and $\emptyset^* = \epsilon$ $\Rightarrow$

5. $r + r = r$

6. $r^*.r^* = r^*$

7. $r.r^* = r^*.r = r^+$

8. $(r^*)^* = r^*$

9. $\epsilon + r.r^* = r^*$

Identities for regular expression

10. $(p + q)^* = (p^*.q^*)^* = (p^* + q^*)^*$

11. $(p.q)^*.p = p.(q.p)^*$

12. $(p+q).r = p.r + q.r$ and $r.(p+q) = r.p + r.q$

1. $r^+ \cup r^* = r^*$

2. $r^+ \cap r^* = r^+$

3. $(r^*)^+ = (r^*)$

4. $(r^+)^* = (r^*)$

5. $((r^*)^+)^* \cdot r^+ = r^+$

6. $(a+b)^* = (a^* + b)^*$

7. $(a+b)^* = (a + b^*)^*$

8. $(a+b)^* \neq (a.b)^*$

9. $(a+b)^* \neq (a^*.b)^*$

10. $(a+b)^* \neq (a.b^*)^*$

Proof the following

$$\checkmark (1+00^*1) + (1+00^*1)(0+10^*1)^*(0+10^*1) = (0^*1(0+10^*1))^*$$

$$(1+00^*1)(\epsilon + (0+10^*1)^*(0+10^*1))$$

$$(1+00^*1)(0+10^*1)^*$$

$$(1 \cdot \epsilon + 00^*1)(0+10^*1)^*$$

$$(\epsilon + 00^*)1(0+10^*1)^*$$

$$\underbrace{\epsilon + 00^*}_{\epsilon + RR^*}$$

$$0^*1(0+10^*1)^*$$

$$\epsilon + R^*R = R^*$$

$$\epsilon + \underbrace{1^*(011)^*}_{R} [\underbrace{1^*(011)^*}_{R^*}]^* = \underbrace{(1+011)^*}_{R^*}$$

$$[1^*(011)^*]^*$$

$$(P+Q)^* = (P^*Q^*)^*$$

$$= (1+011)^*$$

$$R = 1^*(011)^*$$

$$R^* = (1^*(011)^*)^*$$

$$\epsilon + RR^* = R^*$$

$$P = 1$$

$$P^* = 1^*$$

$$Q = (011)$$

$$Q^* = (011)^*$$

$$(P^*Q^*)^* = (1^*(011)^*)^*$$

It states that-

Let P and Q be two regular expressions over Σ .

➤ If P does not contain a null string ϵ , then-

$R = Q + RP$ has a unique solution i.e. $R = QP^*$

NOTE:

➤ Arden's Theorem is popularly used to convert a given DFA to its regular expression.

Proof

$$\begin{aligned} R &= Q + RP \\ R &= Q + QP^*P \\ &\quad Q(\epsilon + P^*P) \\ &\quad QP^* \end{aligned}$$

this is the solution

$$R = QP^*$$

$$\epsilon + R^*R = R^*$$

$$\begin{aligned} R &= Q + RP \\ &= Q + (Q + RP)P \\ &= Q + QP + RP^2 \\ &= Q + QP + (Q + RP)P^2 \\ &= Q + QP + QP^2 + RP^3 \\ &= Q + QP + QP^2 + \dots + QP^n + RP^{n+1} \\ &= Q + QP + QP^2 + \dots + QP^n + QP^*P^{n+1} \\ &= Q(\epsilon + P + P^2 + \dots + P^n + P^*P^{n+1}) \\ &= QP^* \end{aligned}$$

$$R = Q + RP$$

$$\therefore R = QP^*$$

this proof this the unique solution

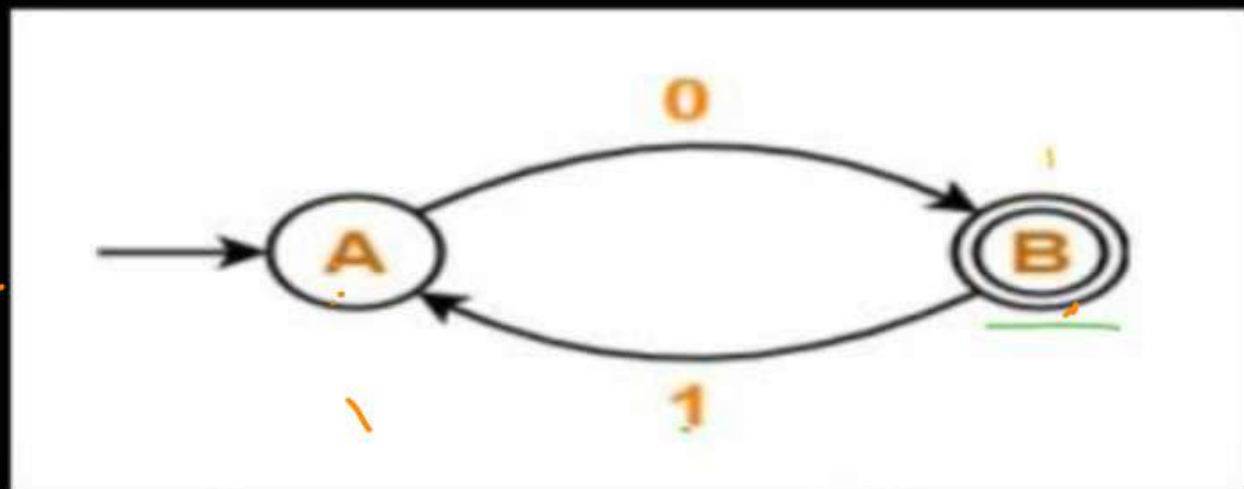
NOTE-

- Arden's Theorem can be used to find a regular expression for both DFA and NFA.
- If there exists multiple final states, then-
 - Write a regular expression for each final state separately.
 - Add all the regular expressions to get the final regular expression.

PROCEDURE TO FIND REGULAR EXPRESSION

- Form an equation for each state considering the transitions which come towards that state.
- Add 'ε' in the equation of initial state.
- Bring final state in the form $R = Q + RP$ to get the required regular expression.

EXAMPLE 1



$$B = 0(10)^*$$

$$A = \epsilon + B \cdot 1 \quad \text{--- (1)}$$

$$B = A \cdot 0 \quad \text{--- (2)}$$

Put A value in eq²

$$B = (\epsilon + B \cdot 1) \cdot 0$$

$$B = \epsilon \cdot 0 + B \cdot 1 \cdot 0$$

$$B = 0 + B \cdot 10$$

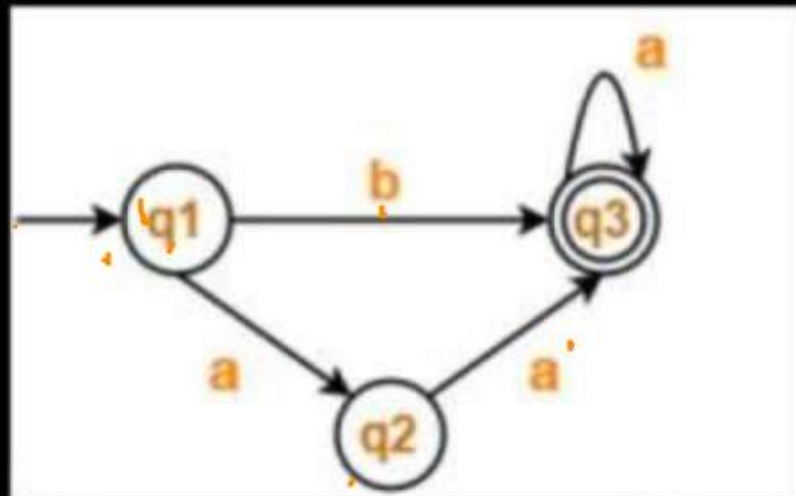
$$R = 0 + R \cdot 10$$

$$R = 0 + R \cdot 10^*$$

$$\epsilon \cdot R = R$$

Gateway classes 7455 9612 84

EXAMPLE 2



$$\begin{aligned}
 q_1 &= \epsilon & \text{--- (1)} \\
 q_2 &= q_1 a & \text{--- (2)} \\
 q_3 &= q_1 b + q_2 a + q_3 a & \text{--- (3)} \\
 &\text{Put } q_1 \text{ value in equation 2} \\
 q_2 &= \epsilon \cdot a = a \\
 \boxed{q_2 = a} & & \text{--- (4)}
 \end{aligned}$$

$$\begin{aligned}
 R &= \emptyset + RP \\
 R &= \emptyset P^*
 \end{aligned}$$

$$q_3 = q_1 \cdot b + q_2 \cdot a + q_3 \cdot a$$

Put q_1 value from eq 1

Put q_2 value from eq 2

$$q_3 = \epsilon \cdot b + a \cdot a + q_3 \cdot a$$

$$b + a \cdot a + q_3 \cdot a$$

$$q_3 = (b + aa) + q_3 \cdot a$$

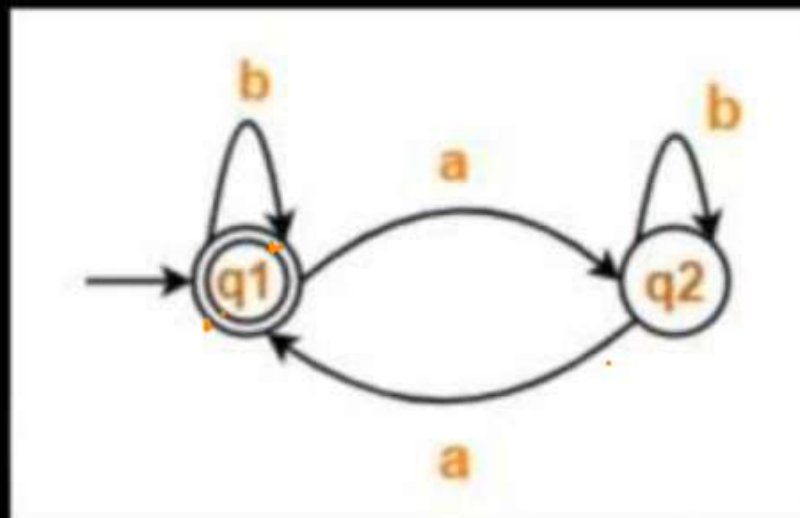
$$R \quad \emptyset \quad + R P$$

$$q_3 = (b + aa) \cdot a^*$$

$$R.E = (b + aa) \cdot a^*$$

$$\begin{aligned}
 \emptyset &= b + aa \\
 P &= a
 \end{aligned}$$

EXAMPLE 3



$$q_1 = \epsilon + q_1 b + q_2 a \quad \text{--- ①}$$

$$q_2 = q_1 a + q_2 b \quad \text{--- ②}$$

$$R = \text{①} + R P$$

$$q_2 = q_1 a b^* \quad \text{--- ③}$$

Put q_2 value in eq 1

$$q_1 = \epsilon + q_1 b + q_2 a$$

$$q_1 = \epsilon + q_1 b + q_1 a b^* a$$

$$R = Q + R P$$

$$q_1 = \epsilon \cdot (b + a b^* a)^*$$

$$q_1 = (b + a b^* a)^*$$

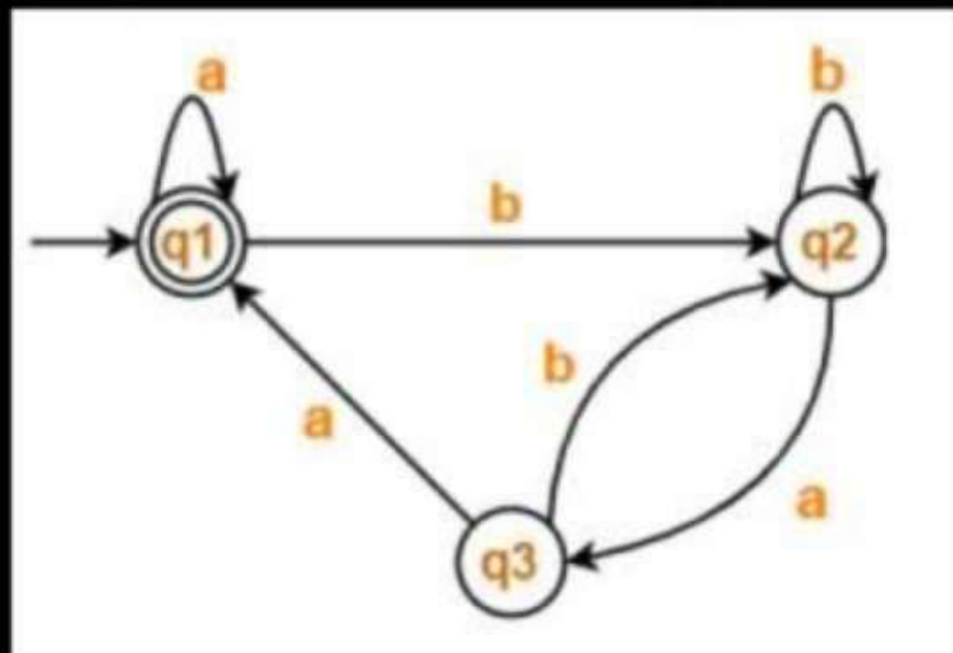
$$RE = (b + a b^* a)^*$$

$$q_2 = q_1 a b^*$$

$$(b) + a b^*$$

$a \cdot \epsilon = a$
 $a \cdot a = a a$
 $a \cdot b b = a b b$
 $a \cdot b b b = a b b b$

EXAMPLE 4



$$q_1 = \epsilon + q_1 a + q_3 a \quad \text{--- (1)}$$

$$q_2 = q_1 b + q_2 b + q_3 b \quad \text{--- (2)}$$

$$q_3 = q_2 a \quad \text{--- (3)}$$

Put eq 3 in eq 2

$$q_2 = q_1 b + q_2 b + q_3 b$$

$$q_1 b + q_2 b + q_2 a b$$

$$q_2 = q_1 b + q_2 (b + ab)$$

$$R = Q + R P$$

$$q_2 = q_1 b (b + ab)^* \quad \text{--- (4)}$$

Put equation 4 in eq 3

$$q_3 = q_2 a$$

$$q_3 = q_1 b (b + ab)^* a \quad \text{--- (5)}$$

Put equation 5 in eq 1

$$q_1 = \epsilon + q_1 a + q_1 b (b + ab)^* a a$$

$$q_1 = \epsilon + (a + b (b + ab)^* aa)$$

$$R = Q + R P$$

$$q_1 = \epsilon (a + b (b + ab)^* aa)^*$$

$$q_1 = (a + b (b + ab)^* aa)^*$$

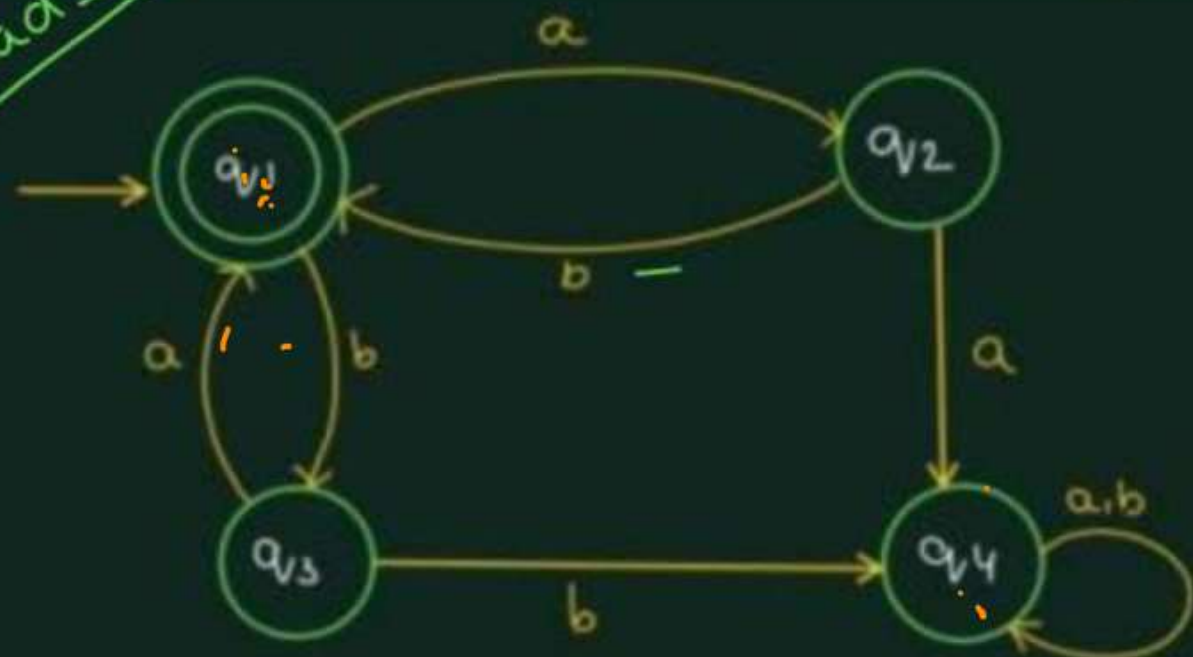
$$RE \quad (a + b (b + ab)^* aa)^*$$

find out the R.E from the
Given

finite
Automata

EXAMPLE 1

Dead state



$$q_1 = \epsilon + q_2 b + q_3 a \quad \text{--- (1)}$$

$$q_2 = q_1 a \quad \text{--- (2)}$$

$$q_3 = q_1 b \quad \text{--- (3)}$$

$$q_4 = q_2 a + q_3 b + q_4 a + q_4 b \quad \text{--- (4)}$$

Put q_2 & q_3 value in equation 1

$$q_1 = \epsilon + q_1 ab + q_1 ba$$

$$q_1 = \epsilon + q_1 (ab + ba)$$

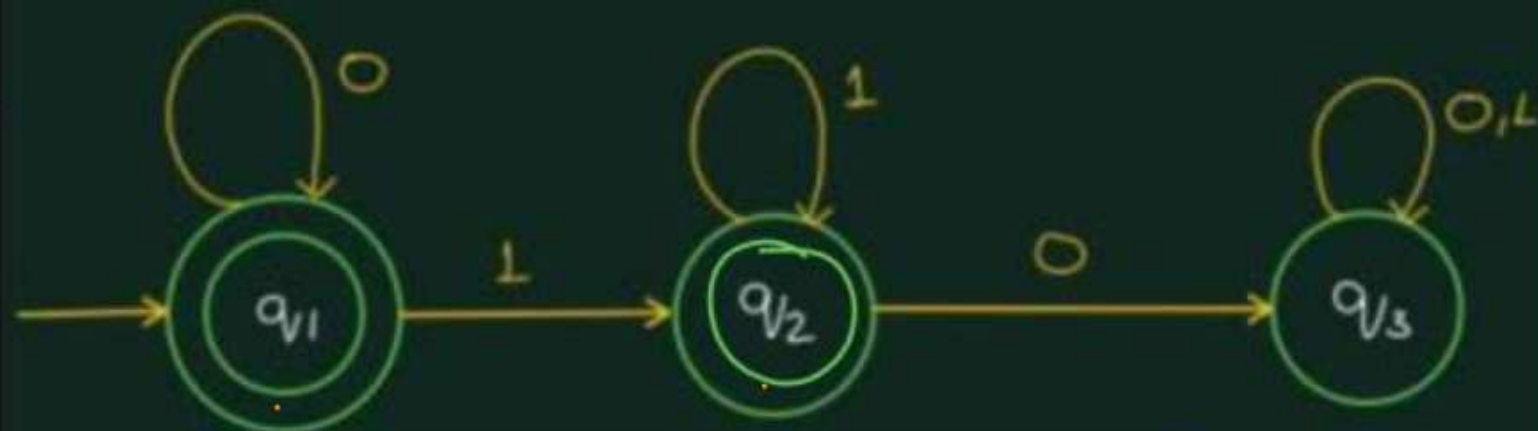
$$R = \emptyset + R \cdot P$$

$$q_1 = \epsilon \cdot (ab + ba)^*$$

$$q_1 = (ab + ba)^*$$

$$\text{R.E} = (ab + ba)^*$$

EXAMPLE 2



$$q_1 = \epsilon + q_1 0 \quad \text{--- ①}$$

$$q_2 = q_1 1 + q_2 1 \quad \text{--- ②}$$

$$q_3 = q_2 0 + q_3 0 + q_3 1 \quad \text{--- ③}$$

if q_1 is my final state

$$q_1 = \epsilon + q_1 0$$

$$R = \emptyset + R P$$

$$q_1 = \epsilon \cdot 0^*$$

$$q_1 = 0^*$$

$$R \cdot E_1 = 0^*$$

if q_1, q_2 are my both final state

$$q_2 = q_1 1 + q_2 1$$

$$q_2 = 0^* 1 + q_2 1$$

$$R = \emptyset + R P$$

$$q_2 = 0^* 1 1^*$$

$$R E_2 = 0^* 1 1^*$$

$$R \cdot E = R E_1 + R E_2$$

$$0^* + 0^* 1 1^*$$

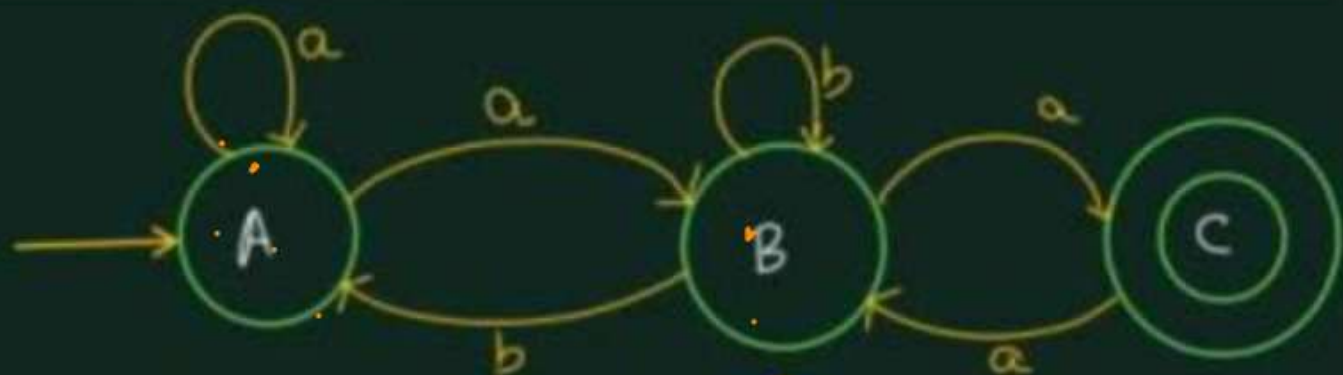
$$0^* (\epsilon + 1 1^*)$$

$$0^* 1^*$$

$$\epsilon + R R^* = R^*$$

$$C + R^* R = R^*$$

EXAMPLE 5



$$A = \epsilon + A \cdot a + B \cdot b \quad \text{--- (1)}$$

$$B = A \cdot a + B \cdot b + C \cdot a \quad \text{--- (2)}$$

$$C = B \cdot a \quad \text{--- (3)}$$

Put eq 3 in eq 2

$$B = A \cdot a + B \cdot b + C \cdot a$$

$$B = A \cdot a + B \cdot b + B \cdot a a$$

$$B = A \cdot a + B(b + aa)$$

$$R = \emptyset + R P$$

$$B = A a (b + aa)^* \quad \text{--- (4)}$$

Put eq 4 in eq 1

$$A = \epsilon + A \cdot a + B \cdot b$$

$$A = \epsilon + A \cdot a + A a (b + aa)^* b$$

$$A = \epsilon + A (a + a (b + aa)^* b)$$

$$A = \epsilon \cdot (a + a (b + aa)^* b)^* \quad \text{--- (5)}$$

$$A = (a + a (b + aa)^* b)^* \quad \text{--- (6)}$$

Put eq 6 in eq 4

$$B = (a + a (b + aa)^* b)^* a (b + aa)^* \quad \text{--- (7)}$$

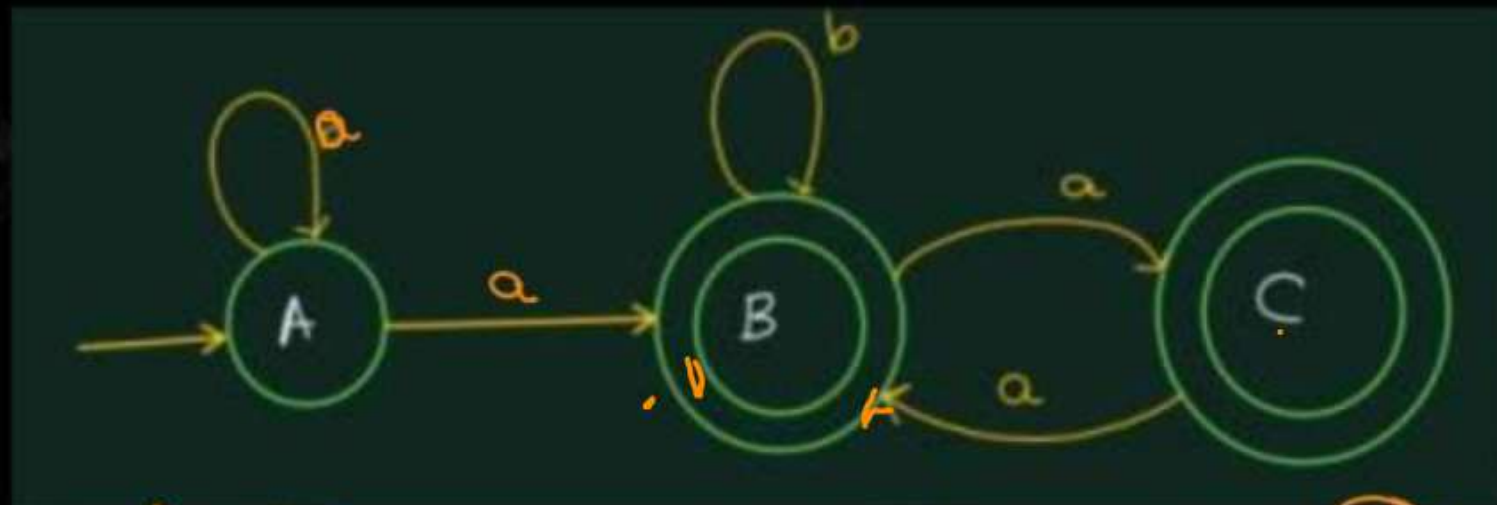
Put eq 7 in eq 3

$$C = B \cdot a$$

$$(a + a (b + aa)^* b)^* a (b + aa)^* a$$

↓
R.E

EXAMPLE 6



$$R = \emptyset + RP$$

$$A = \epsilon + A \cdot a \quad \text{--- (1)}$$

$$B = A \cdot a + B \cdot b + C \cdot a \quad \text{--- (2)}$$

$$C = B \cdot a \quad \text{--- (3)}$$

Put eq 3 in eq 2

$$B = A \cdot a + B \cdot b + B \cdot aa \rightarrow B = A \cdot a + B(b + aa)$$

$$R = \emptyset + RP$$

$$A = \epsilon \cdot a^* \quad A = a^* \quad \text{--- (5)}$$

$$B = A \cdot a(b + aa)^* \quad \text{--- (6)}$$

Put eq 5 in eq 6

$$B = a^* a(b + aa)^* \quad \text{--- (7)}$$

Put equation 7 in eq 3

$$C = B \cdot a$$

$$C = a^* a(b + aa)^* a \quad \text{--- (8)}$$

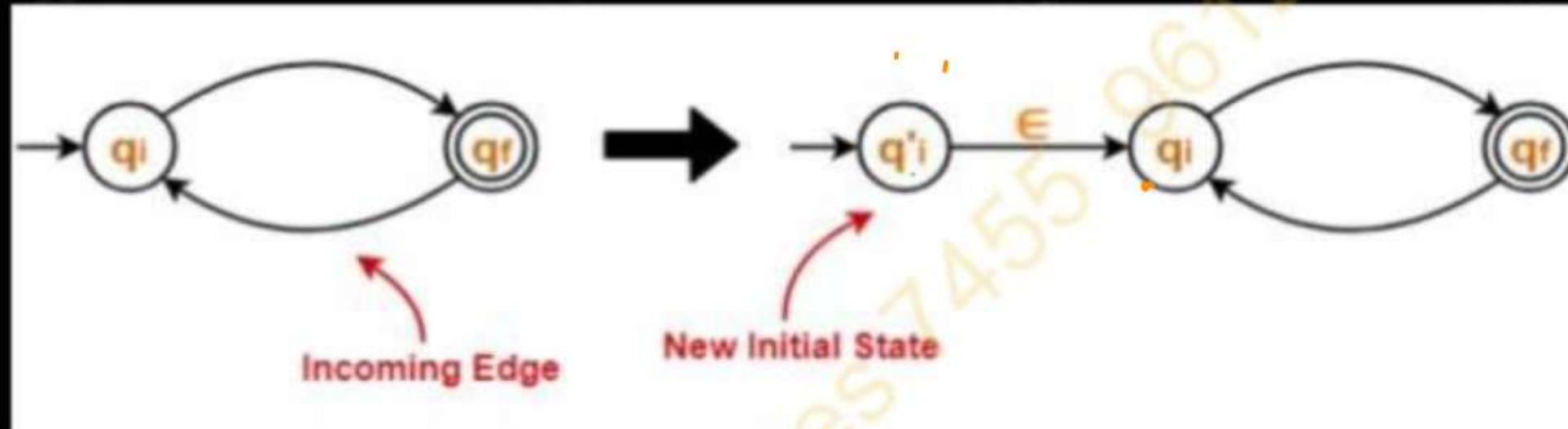
$$R.E = B + C$$

$$a^* a(b + aa)^* + a^* a(b + aa)^* a$$

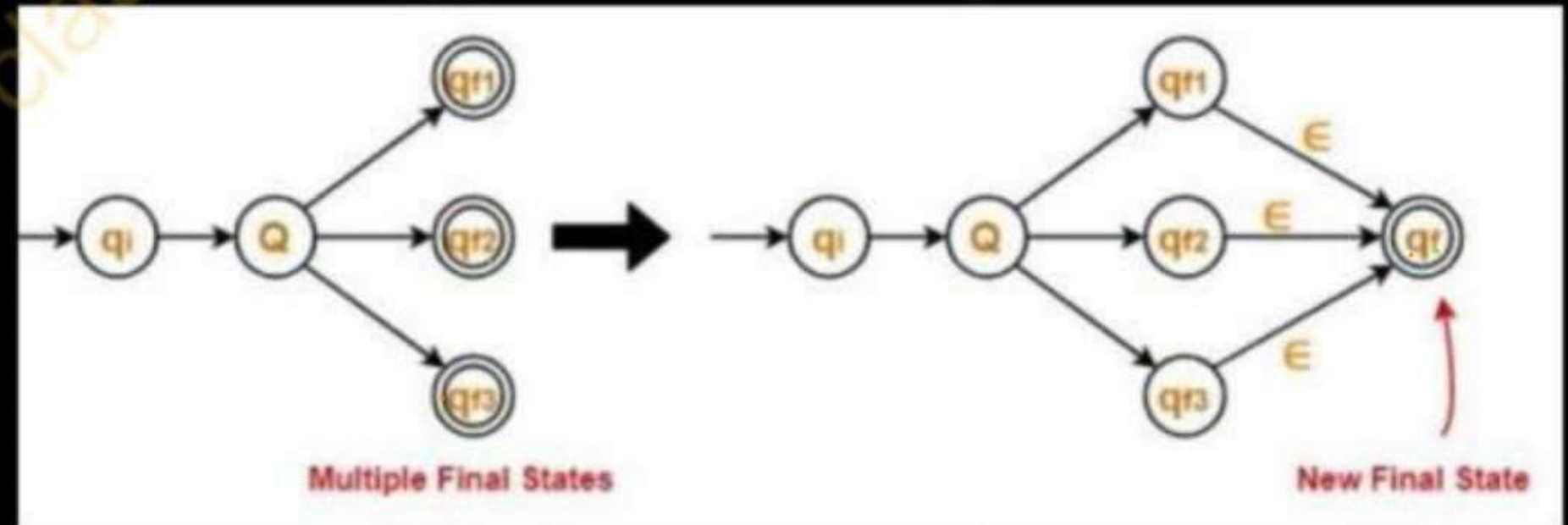
$$= a^* a(b + aa)^* (\epsilon + a)$$

State elimination method

1. The initial state of the ^{NFA}DFA must not have any incoming edge.

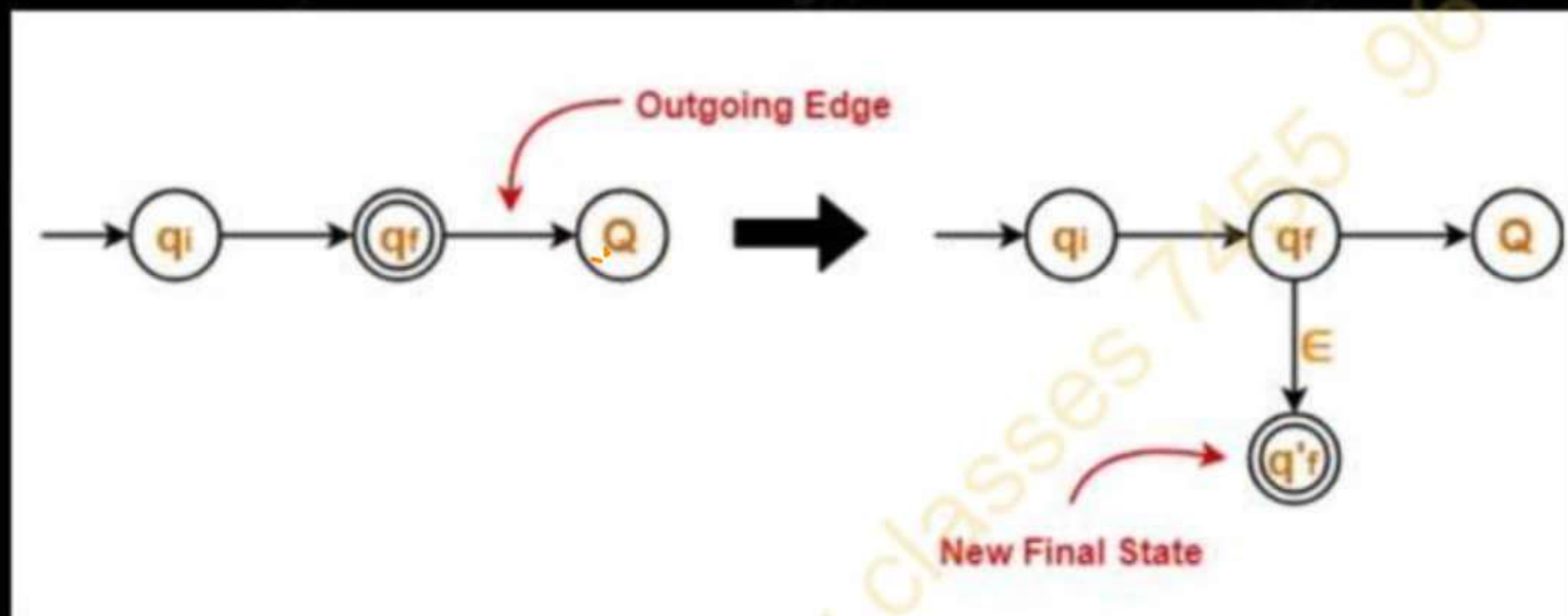


2. There must exist only one final state in the DFA.

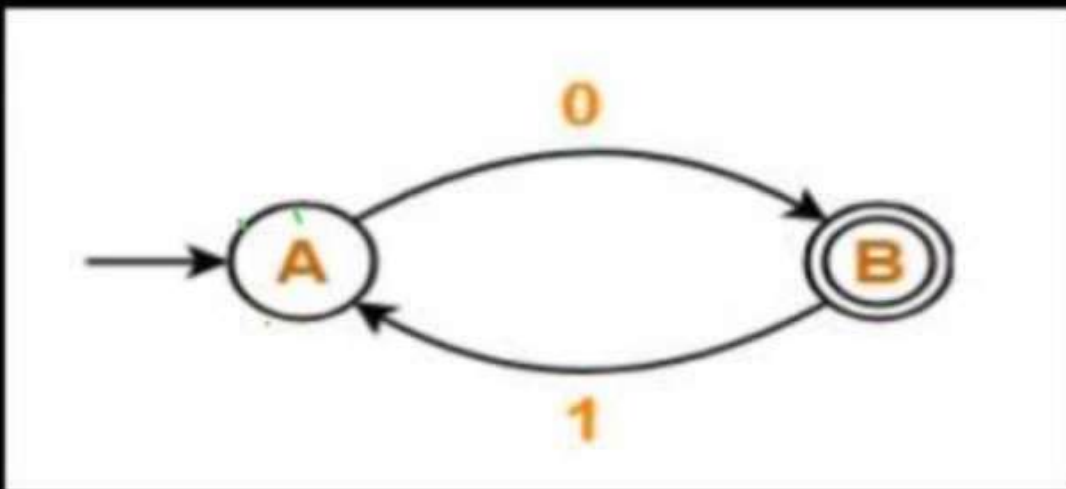


State elimination method

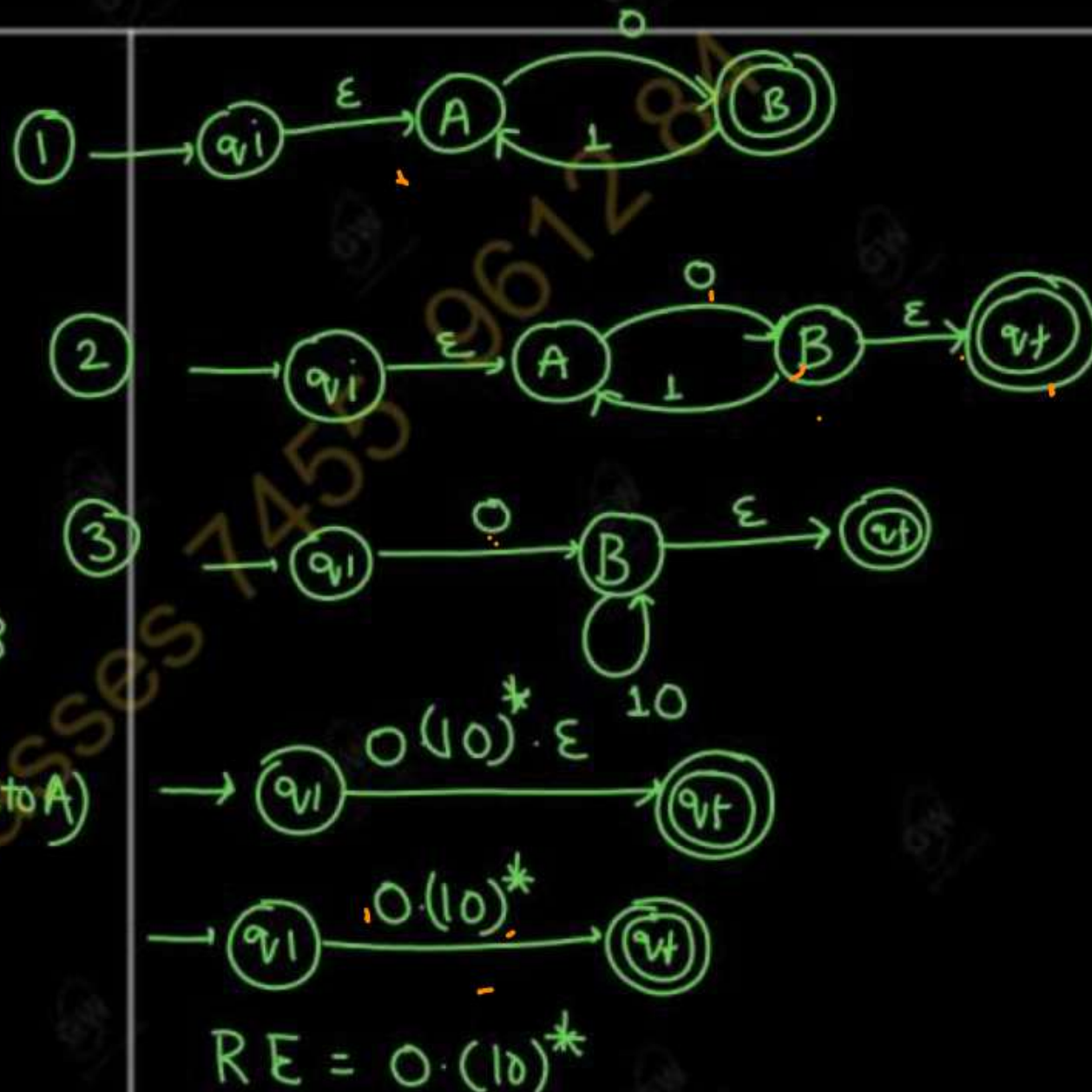
3. The final state of the DFA must not have any outgoing edge



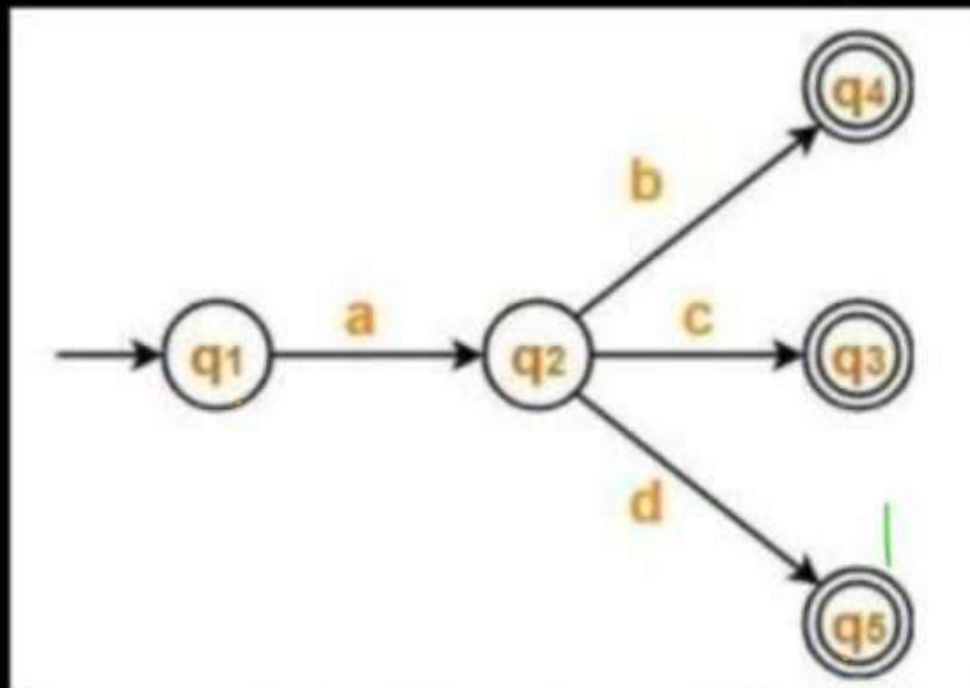
EXAMPLE 1



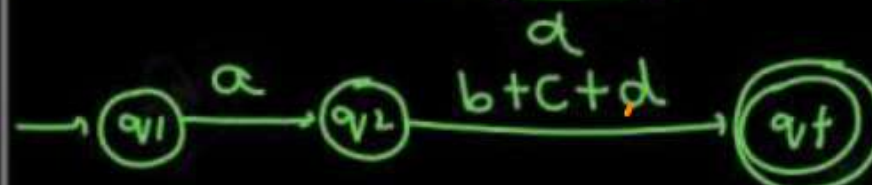
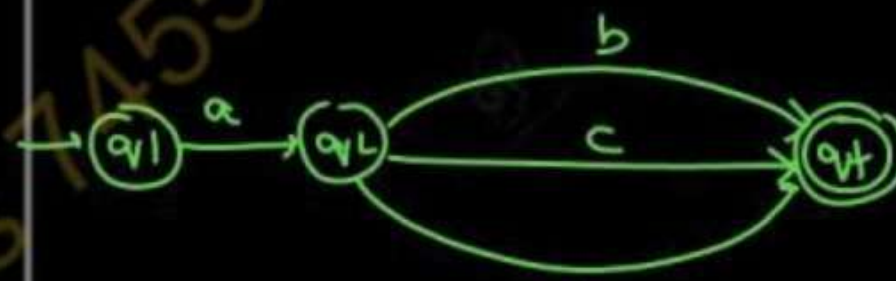
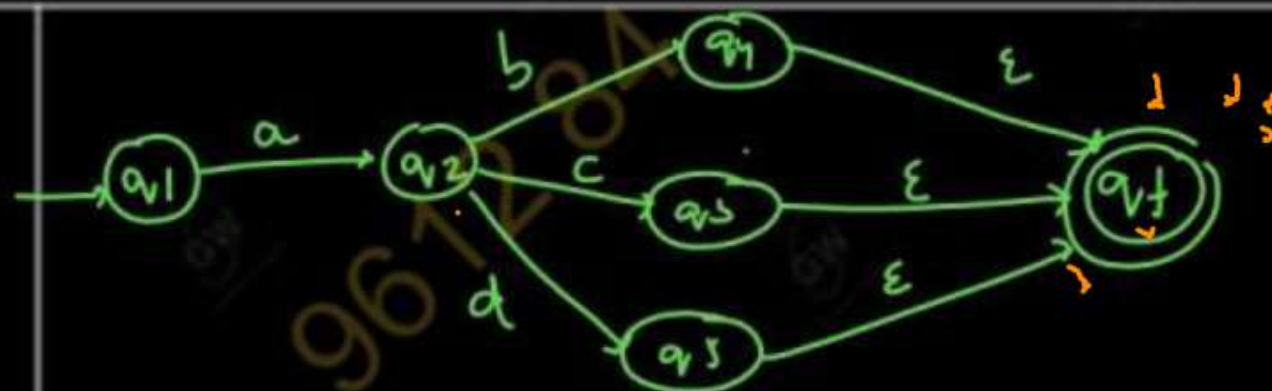
- ① there is an incoming (initial state) (A to B)
- ② Single final state ✓
- ③ there is an outgoing edge from final state (B to A)



EXAMPLE 2

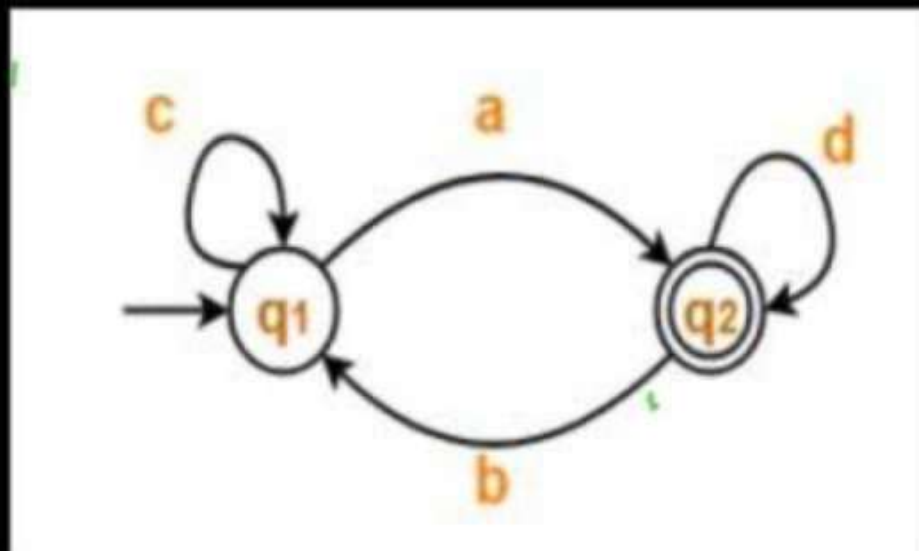


- NO incoming edge on initial state ✓
- Multiple final state
- NO outgoing edge from final state

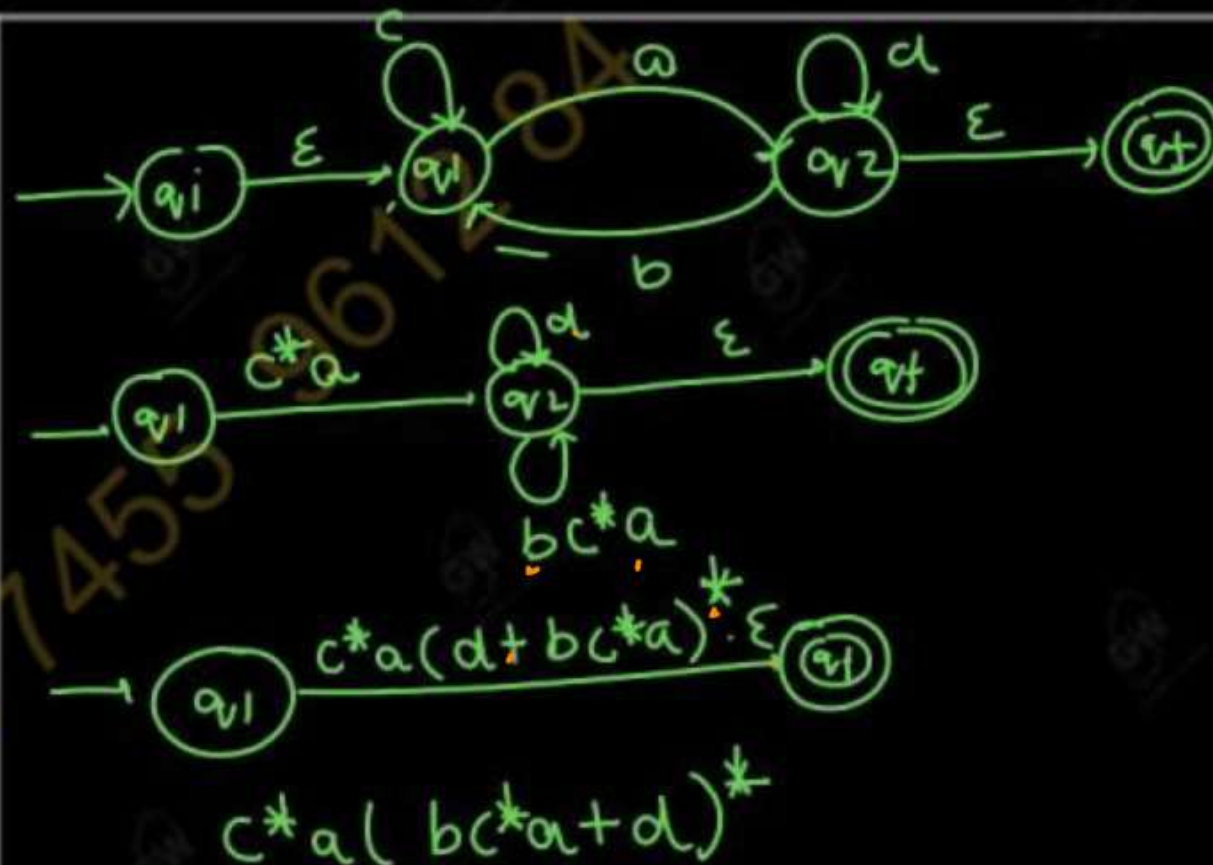


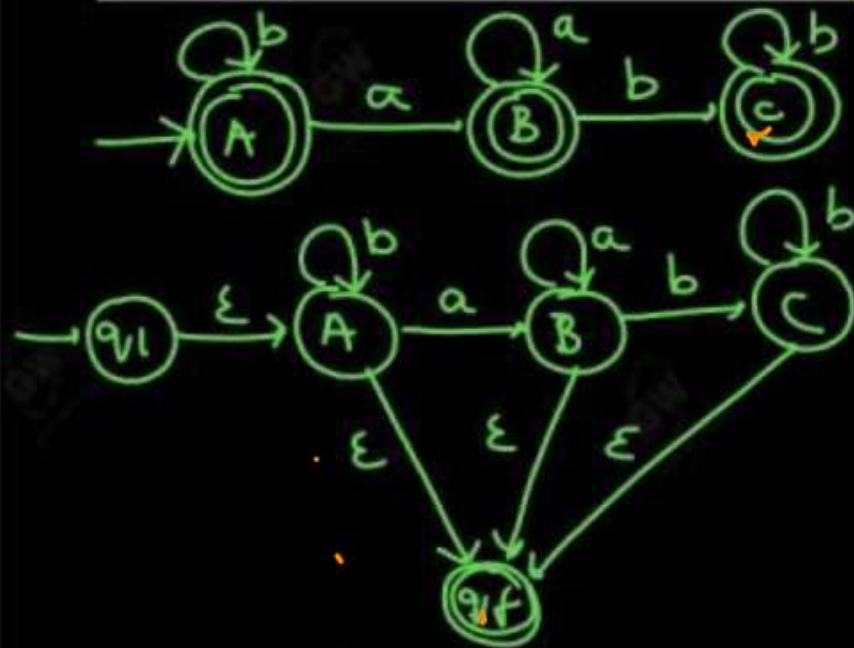
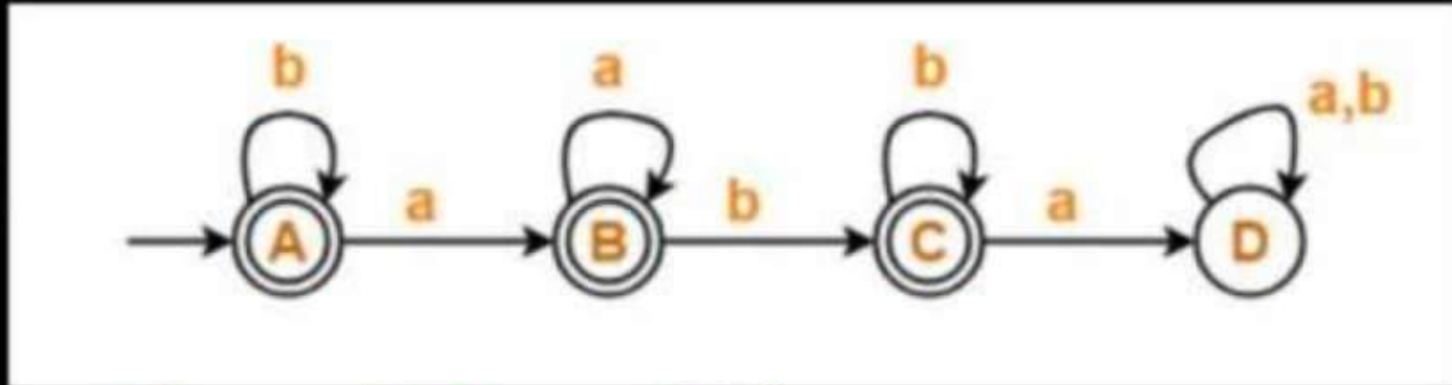
R.E. $a \cdot (b+c+d)$

EXAMPLE 3

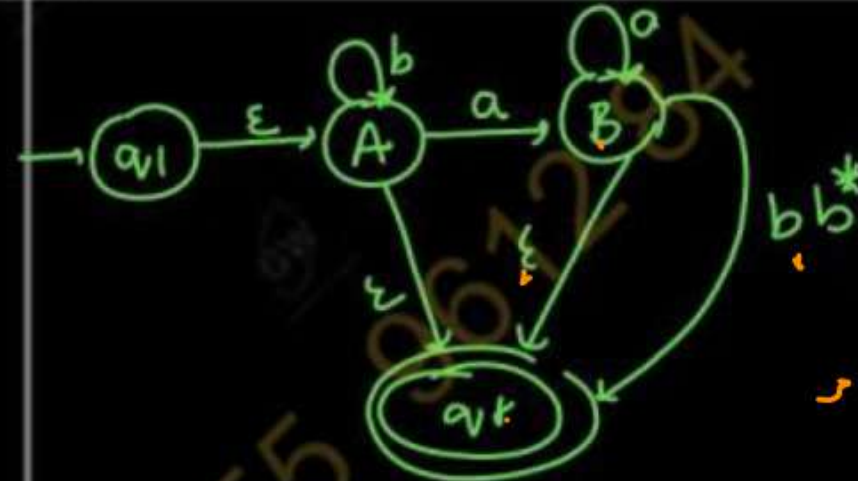


- Incoming in entered state
- Single final state
- Exiting edge from final state





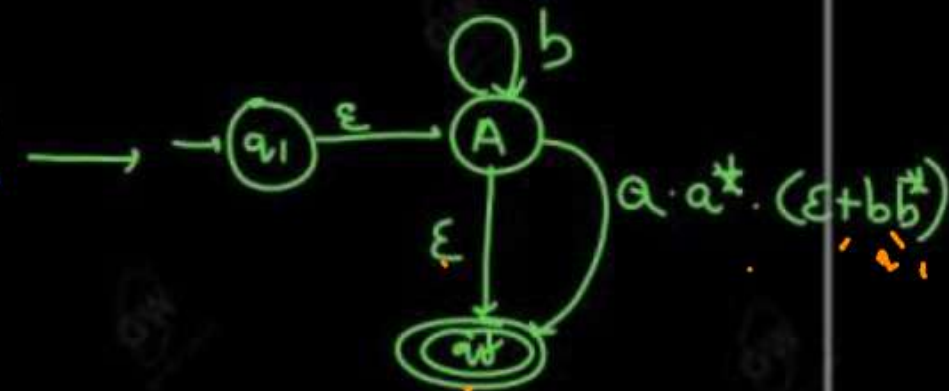
$$b^* \cdot \epsilon = b b^*$$



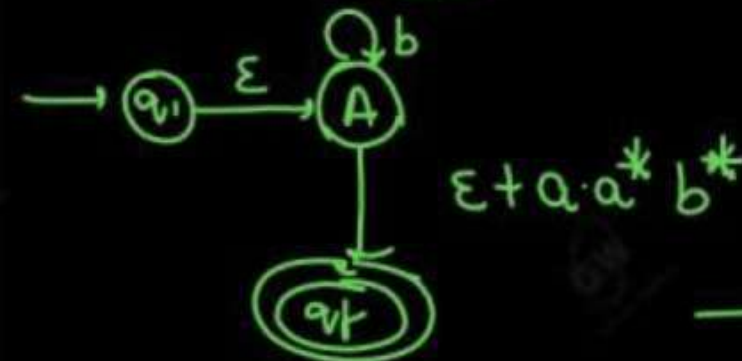
$$\epsilon + R R^* = R^*$$



$$\epsilon + b b^* = b^*$$



$$a \cdot a^* (\epsilon + b b^*)$$

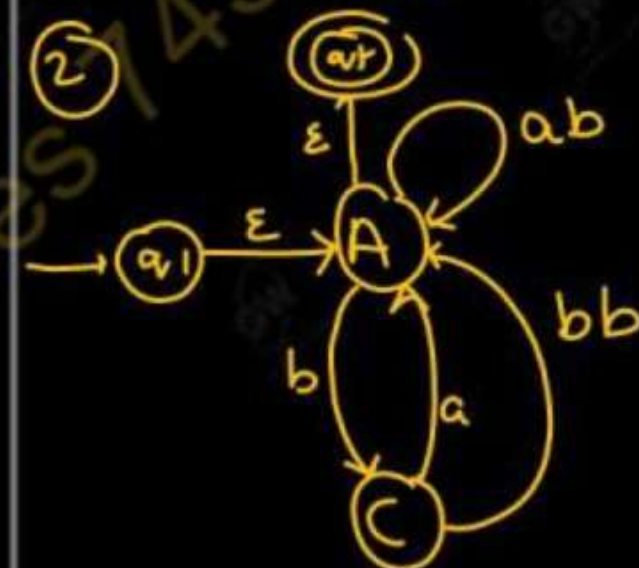
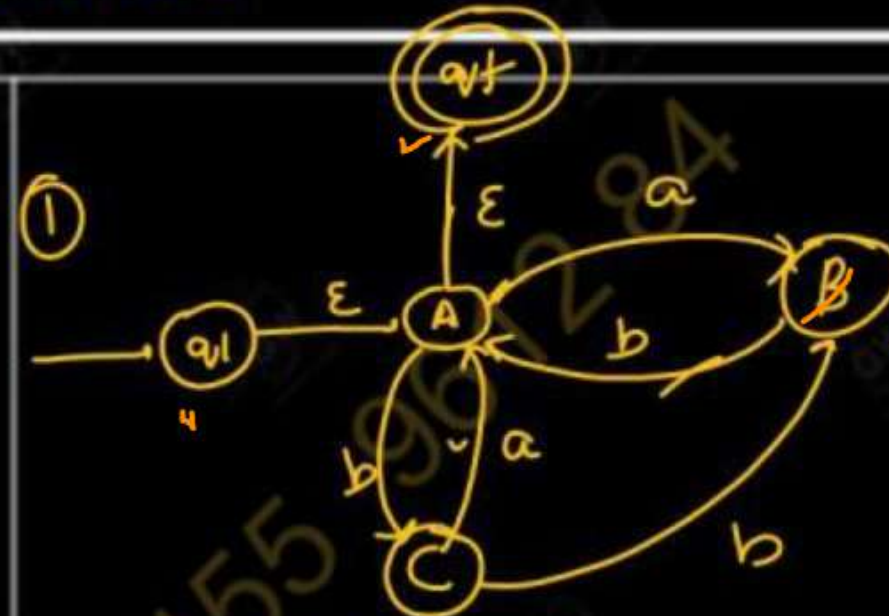
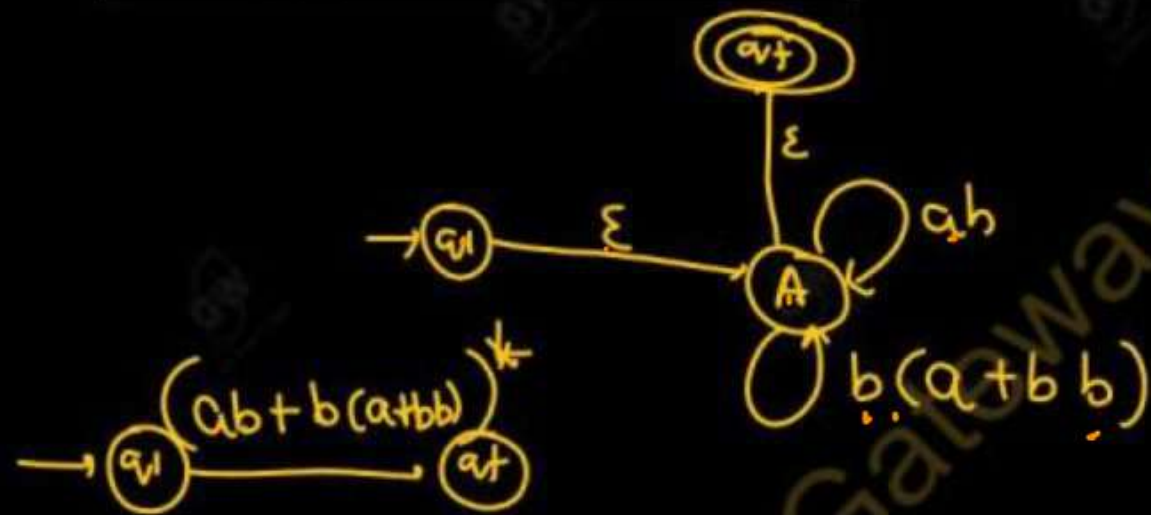
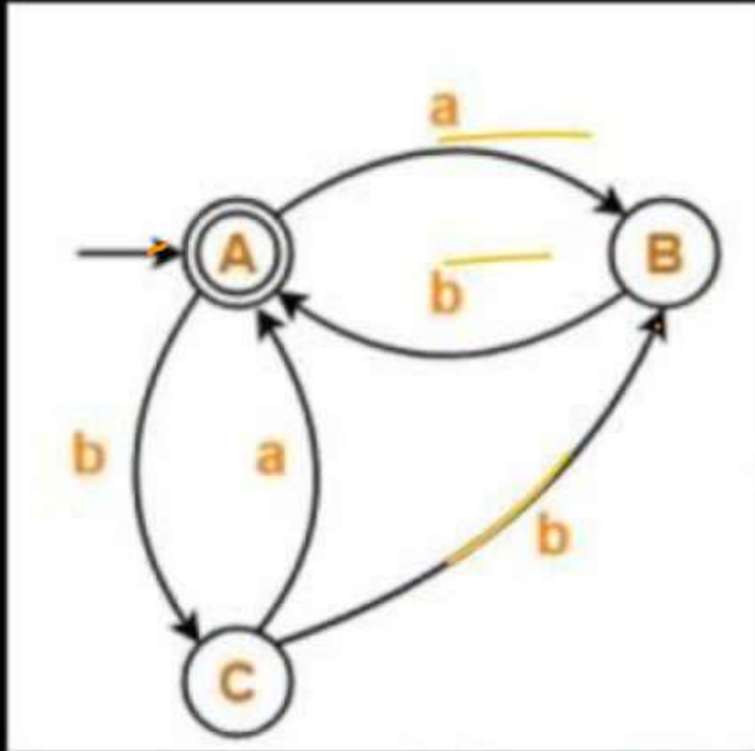


$$\epsilon + a \cdot a^* b^*$$



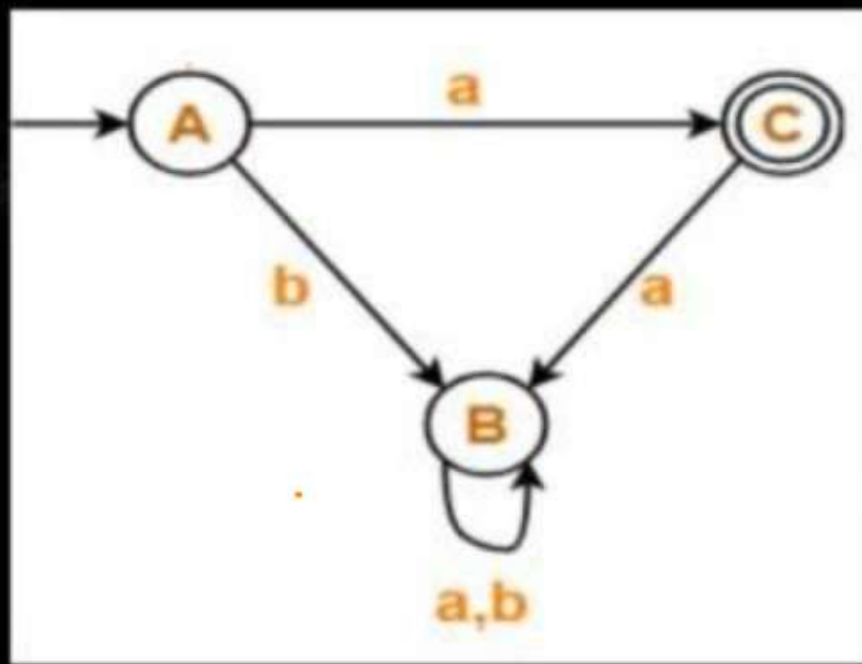
$$RE = b^* (\epsilon + a a^* b^*)$$

EXAMPLE 5



$$RE = (ab + b(a+bbb))^*$$

EXAMPLE 6



R.E = a

The language accepted by **Finite AUTOMATA** is known as **Regular Language**. Pumping Lemma is used to prove that a Language is not Regular. It cannot be used to prove that a language is Regular.

What is Pumping Lemma?

The term Pumping Lemma is made up of two words:-

Pumping: The word pumping refers to generating many input strings by pushing a symbol in an input string repeatedly.

Lemma: The word Lemma refers to the intermediate theorem in a proof.

There are two Pumping Lemmas, that are defined for

- Regular Languages
- Context-Free Languages

Pumping Lemma For Regular Languages

Theorem: If A is a Regular Language, then A has a Pumping Length ' P ' such that any string ' S ' where $|S| \geq P$ may be divided into three parts $S = xyz$ such that the following conditions must be true:

1.) $xy^iz \in A$ for every $i \geq 0$

2.) $|y| > 0$

3.) $|xy| \leq P$

In simple words, if a string y is 'pumped' or insert any number of times, the resultant string still remains in A

$L = \{a^n b^n \mid n \geq 1\}$ is not a regular language

Let L is Regular language

$L = \{ab, aabb, aaabbb, \dots\}$

$P = 4$

$|S| \geq P$

$S = aabb$

$|S| = 4$

$|S| \geq P \quad |4| \geq 4 - \text{True case 1}$

$S = xyz$

$aabb$

$x \quad y \quad z$

$y = a \quad |y| = 1 \quad |y| > 0$

$xy = aa \quad |xy| = 2 \quad 2 > 0 \text{ T.}$

$|xy| \leq P$

$xy^iz \in L \quad i \geq 0 \quad 2 \leq 4 = T$

$a(a)(bb)$

$i = 0 \quad a \cdot \epsilon \cdot bb = abb \notin L$

$i = 1 \quad a \cdot a \cdot bb \in L$

$i = 2 \quad a \cdot aa \cdot bb \notin L$

this not Regular Language

$S = ab$
 $|S| = 2$
 $|S| \geq P$
 $2 \geq 4 - \text{False}$

$S = aabb$

if a string y is pumped number of time string will not remain in language

Pumping Lemma For Regular Languages

$L = \{a^n b^{2n} \mid n \geq 1\}$ is not a regular language

let L is a Regular Language

$L = \{abb, aabbbb, aaaa bbbb b \dots\}$

$P = 6$ (let)

$S = abb$

$|S| = 3$

$|S| > P$

$3 > 6$ - False

$S = aabbbb$

$|S| = 6$

$|S| \geq 1$

$6 \geq 6$ T

$S = \underbrace{aa}_x \underbrace{bbb}_y \underbrace{b}_z$

① $|y| > 0$ $y = bb$

$|y| = 2$

$2 > 0$ T

$|xy| \leq P$

② $xy = aabbb$

$|xy| = |aabbb| = 5$

$5 \leq P$ T

$5 \leq 6$ T

$S = aabcb^i b$

$i=0$ $aab \in b = aabb \notin L$

$i=1$ $aabbbb \in L$

$i=2$ $aabbbbbbb \notin L$

this not a Regular language

Case II

$S = \underbrace{aa}_x \underbrace{bbb}_y \underbrace{b}_z$

$x = a$ $|y| = 3$

$y = abb$ $|xy| = 4$

$z = b$

① $|y| > 0$ $3 > 0$ T

② $|xy| \leq P$ $4 \leq 6$ T

③ $S = xy^i z \in L \ i \geq 0$

$S = a(aab)^i b$

$i=0$ $a \in bb = aabb \in L$

$i=1$ $aabbbb \in L$

$i=2$ $aabbbabbb \notin L$

from these two cases
we proof this not
a Regular language

Pumping Lemma For Regular Languages

$L = \{a^n \mid n \text{ is prime}\}$ is not a regular language

$n = 2, 3, 5, 7, \dots$

Let L is a regular language

$L = \{aa, aaa, aaaaa, \dots\}$
 $p = 3$

$s = aaaaa$

$|s| = 5$

$|s| \geq p \quad s \geq 3$
 True

$s = \underbrace{a}_{x} \underbrace{a}_{y} \underbrace{aaa}_{z}$

$x = a$

$y = a$

$z = aaa$

① $|y| > 0$

$|a| > 0$

$1 > 0 \quad T$

②

$|xy| \leq p$

$|aa| \leq 3$

$2 \leq 3 \quad T$

$s = xyz \in L \quad i \geq 0$

$s = a(a)^i aaa$

$i = 0 \quad a \in aaa = aaaa \notin L$

→ This not Regular

Case 2

$s = \underbrace{aa}_x \underbrace{aa}_y \underbrace{a}_z$

①

$|y| > 0$

$|a| > 0$

$1 > 0 \quad T$

$x = aa$

$y = a$

$z = a$

②

$|xy| \leq p$

$|aa| \leq 3$

$3 \leq 3 \quad T$

$s = xy^i z \in L \quad i \geq 0$

$s = aa(a)^i aa$

$i = 0 \quad aa \in aa \quad aaaa \notin L$

$i = 1 \quad aaaaa \in L$

$i = 2 \quad aa.aaa \notin L$

This not
a Regular
One

Pumping Lemma For Regular Languages

$L = \{a^{n^2} \mid n \geq 0\}$ is not a regular language

$$L = \{a^0, a^1, a^4, a^9, a^{16}, \dots\}$$

$$L = \{\epsilon, a, a^4, a^9, a^{16}, \dots\}$$

Let L is a Regular Language

$$P = 3$$

$$s = aaaa$$

$$|s| \geq P$$

$$|aaaa| \geq 3$$

$$4 \geq 3$$

true

$$s = \overset{x}{a} \overset{y}{a} \overset{z}{aa}$$

$$|x| > 0 \quad |a| > 0 \quad |z| > 0 \quad T$$

$$|xy| \leq P \quad |aaa| \leq P \quad 2 \leq 3 \quad T$$

$$s = xyz \in L \quad i \geq 0$$

$$s = a(a)aa$$

$$i=0 \quad a \in L \quad aa \notin L \quad aaa \notin L$$

this not a regular language

$$s = \overset{x}{a} \overset{y}{a} \overset{z}{aa}$$

①

$$|x| > 0 \quad |a| > 0 \quad |z| > 0 \quad \text{true}$$

②

$$|xy| \leq P \quad |aaa| \leq 3 \quad 3 \leq 3 \quad \text{true}$$

$$s = xyz \in L \quad i \geq 0$$

$$a(a)a$$

$$i=0 \quad aa \in L \quad a \notin L$$

$$i=1 \quad aaaa \in L$$

$$i=2 \quad aaaaaa \notin L$$

from these two case we confirmed that L is not a regular language

Recursive definition for a regular expression

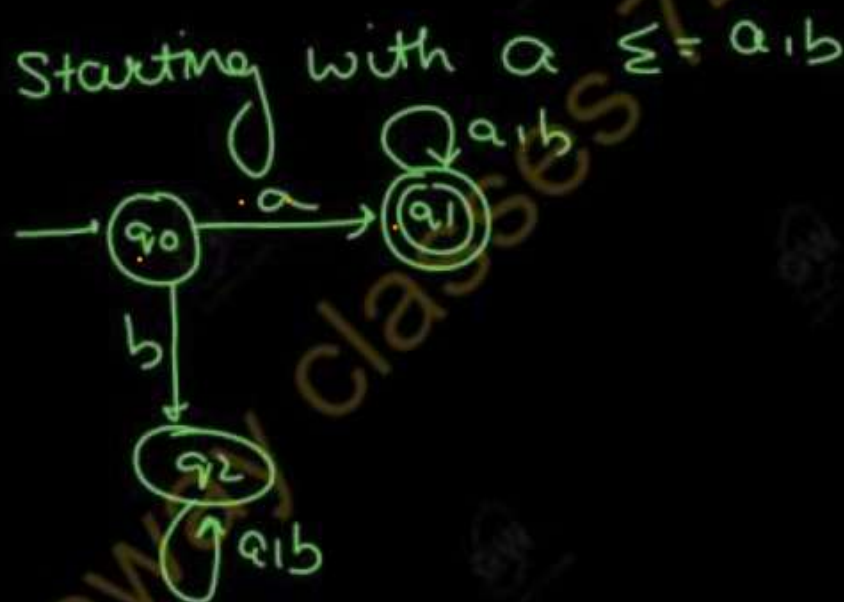
A Regular Expression can be recursively defined as follows –

- ϵ is a Regular Expression indicates the language containing an empty string. ($L(\epsilon) = \{\epsilon\}$)
- ϕ is a Regular Expression denoting an empty language. ($L(\phi) = \{\}$)
- x is a Regular Expression where $L = \{x\}$
- If X is a Regular Expression denoting the language $L(X)$ and Y is a Regular Expression denoting the language $L(Y)$, then
 - ❑ $X + Y$ is a Regular Expression corresponding to the language $L(X) \cup L(Y)$ where $L(X+Y) = L(X) \cup L(Y)$.
 - ❑ $X \cdot Y$ is a Regular Expression corresponding to the language $L(X) \cdot L(Y)$ where $L(X \cdot Y) = L(X) \cdot L(Y)$
 - ❑ R^* is a Regular Expression corresponding to the language $L(R^*)$ where $L(R^*) = (L(R))^*$

$R = ab$
 $R^* = (ab)^*$

If we apply any of the rules several times from 1 to 5, they are Regular Expressions.

- Transition diagram is a special kind of flowchart for language analysis. In transition diagram the boxes of flowchart are drawn as circle and called as states. States are connected by arrows called as edges. The label or weight on edge indicates the input character that can appear after that state.
- It is a graph which consists of series of state and there is successful path begin at start state end at the final state



APPLICATION OF PUMPING LEMMA

➤ to prove that a particular language is non-regular

Let L is Regular Language

$$L = \{a^1, a^8, a^{27}, a^{64}, \dots\}$$

Let $p = 7$

$S = a$	$S = aaaaaaaaa$
$ S = 1$	$ S = 9$
$ S \geq p$	$ S > p$
$1 > 7$	$9 > 7$
(False)	T

case 1

$$S = \underbrace{aa}_x \underbrace{aaa}_y \underbrace{aaa}_z$$

1. $ y > 0$	2. $ xy \leq p$
$y = aaa$	$ aaaaa \leq 7$
$ y = 3$	$5 \leq 7$
$3 > 0$ T	T

3. $S = xy^i z \in L, i \geq 0$

$$aa(aaa)^i(aaa)$$

$i=0 \quad aa \in aaa = a^5 \notin L$

this is not Regular lang.

(7 marks)
Prove the language $L = \{a^n \mid n \text{ is a perfect cube}\}$ is not a regular language

case 2

$$S = \underbrace{aaaa}_x \underbrace{a}_y \underbrace{aaa}_z$$

1. $ y > 0$	2. $ xy \leq p$
$ a > 0$	$ aaaaa \leq 7$
$1 > 0$	$5 \leq 7$ T
T	

3. $S = xy^i z \in L, i \geq 0$

$$aaaa(a)^i aaa$$

$i=0 \quad aaaa \in aas = a^7 \notin L$

$i=1 \quad aaaaaaaaa = a^8 \in L$ ✓

$i=2 \quad aaaa aaaaaaaa = a^9 \notin L$

this is not Regular Lang.

case 3

$$S = \underbrace{aaa}_x \underbrace{aaa}_y \underbrace{aa}_z$$

1. $ y > 0$	2. $ xy \leq p$
$ aaa > 0$	$ aaaaaa \leq 7$
$3 > 0$	$6 \leq 7$ T
T	

3. $S = xy^i z \in L, i \geq 0$

$$aaa(aaa)^i aa$$

$i=0 \quad aaa \in aa = a^5 \notin L$

$i=1 \quad aaaa aaaa = a^8 \in L$ ✓

$i=2 \quad aaa aaaaaaaa = a^{11} \notin L$

- If a language can be expressed by finite automata, transition graph or regular expression, then it can also be expressed by other two as well
- It may be noted that theorem is proved by proving the three following part-

➤ KLEENE THEOREM PART 1

If a language is accepted by finite automata then it must be accepted by transition graph

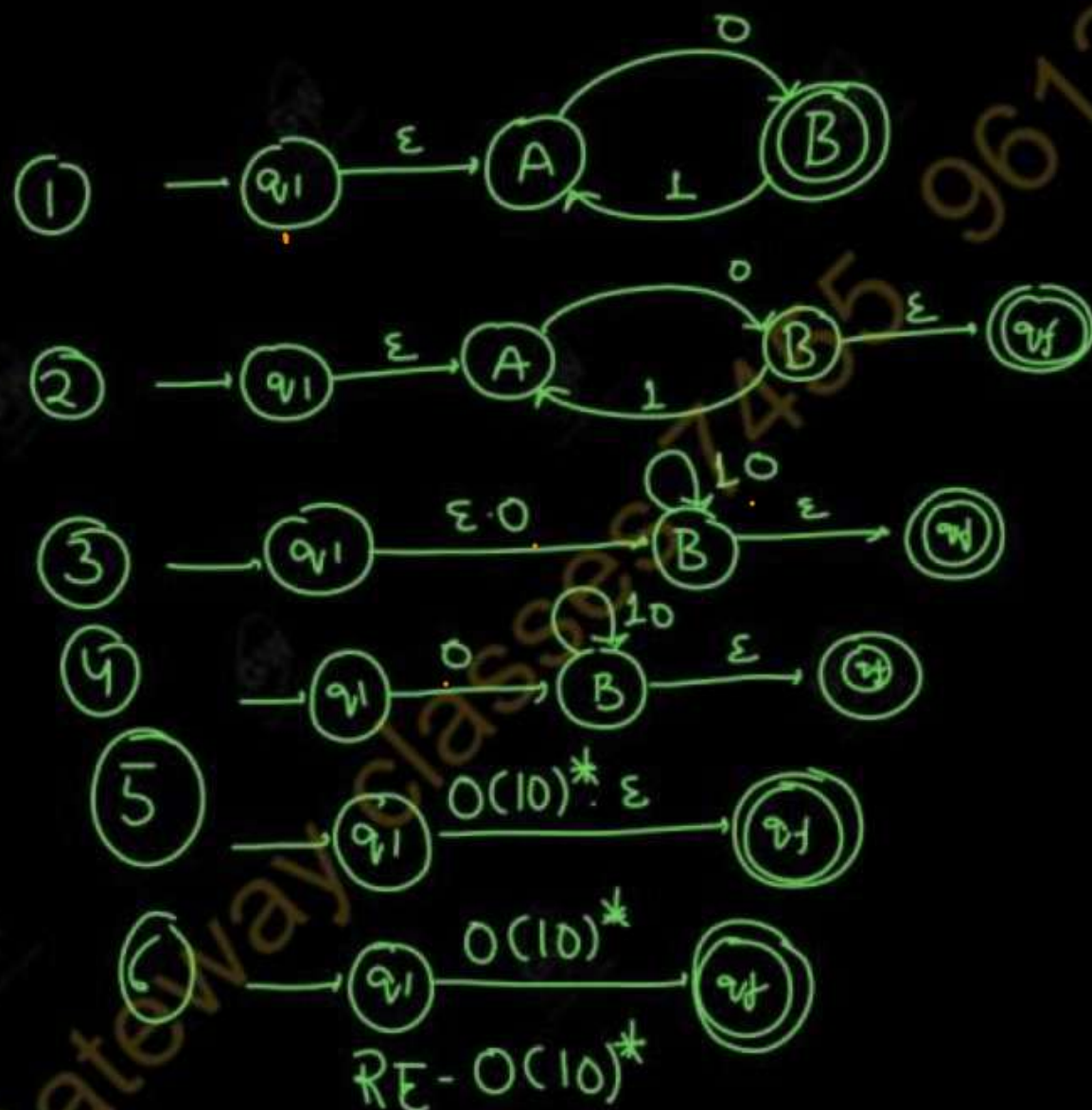
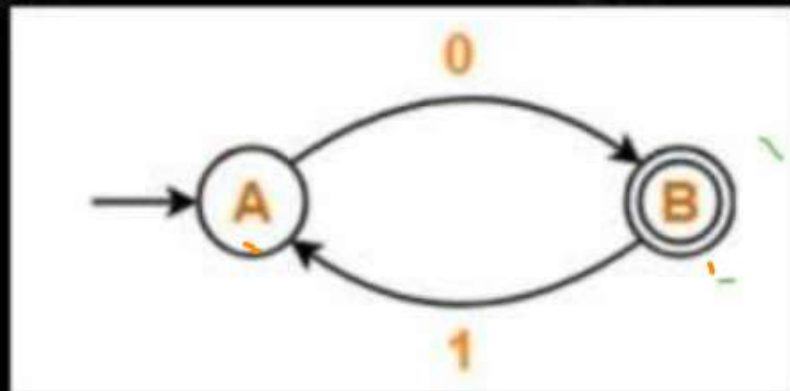
PROOF: every finite automata is considered to be the transition graph then nothing to proof

➤ KLEENE THEOREM PART 2

If a language is accepted by transition graph then it must be expressed by regular expression.

PROOF

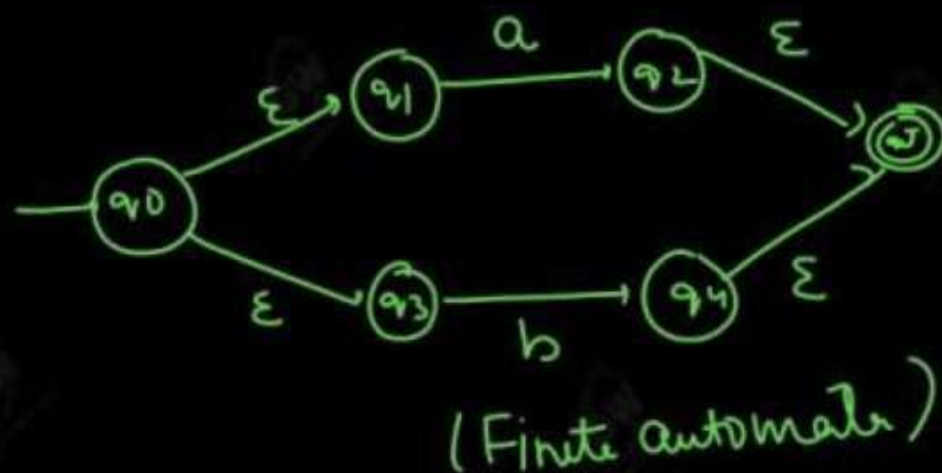
Arden's theorem



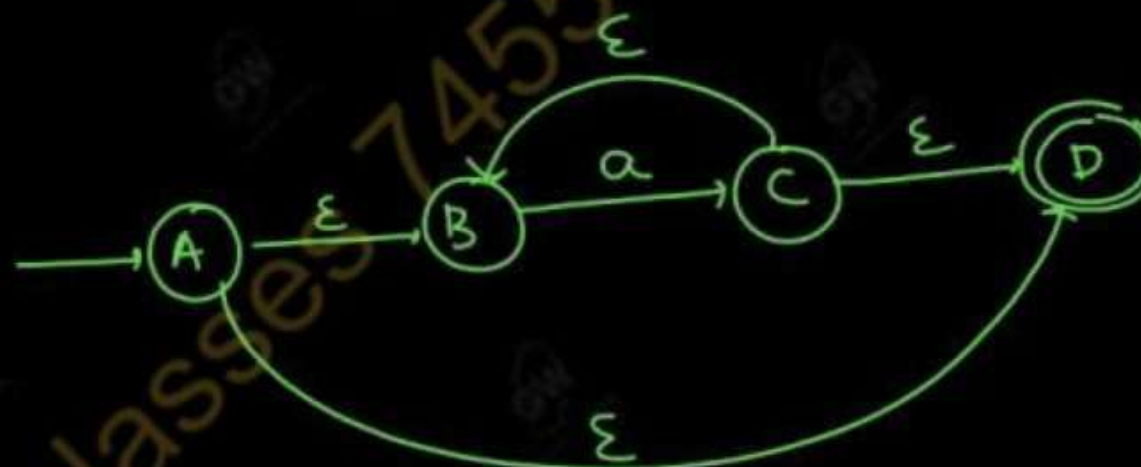
KLEENE THEOREM PART 3

If a language is expressed by a regular expression then it can be accepted by a finite automata.

R.E = $a+b$



a^*
 $L = \{\epsilon, a, aa, aaa, \dots\}$



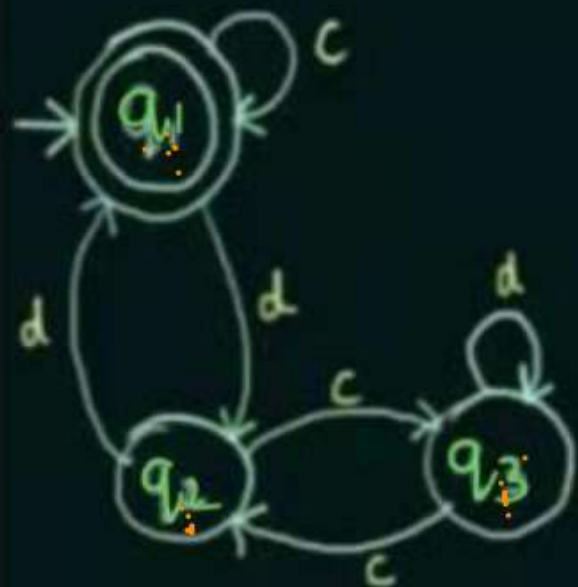
Equivalence of two finite automata

1. For any pair of states $\{q_i, q_j\}$ the transition for input a belong Σ is defined by $\{q_a, q_b\}$ where $\delta\{q_i, a\}=q_a$ and $\delta\{q_j, a\}=q_b$ the two automata are not equivalent if for a pair $\{q_a, q_b\}$ one is intermediate state and other is final state
2. if initial state is final state of one automation then in second automation also initial state must be the final state for them to be equivalent

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Equivalence of two finite automata

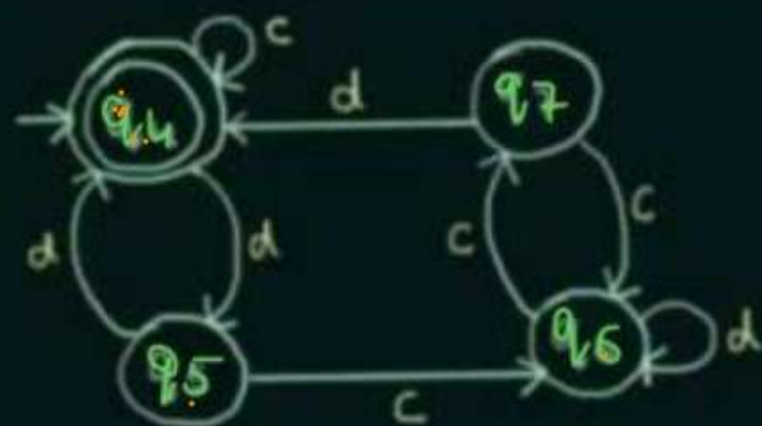
Initial state of automata A is q_1 & final state is q_1
 Initial state of automata B is q_4 & final state is q_4 $\Rightarrow$ condⁿ ✓



A

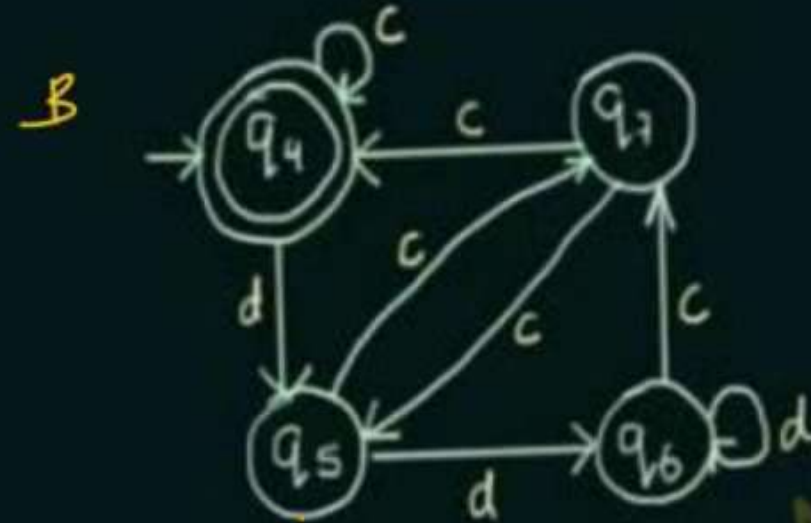
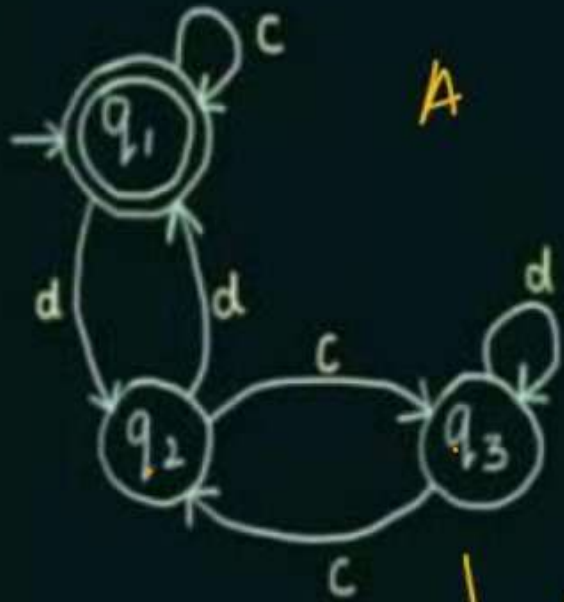
	c	d
$\{q_1, q_4\}$	$\{q_1, q_4\}$ FS FS	$\{q_2, q_5\}$ IS IS
$\{q_2, q_5\}$	$\{q_3, q_6\}$ IS IS	$\{q_1, q_4\}$ FS FS
$\{q_3, q_6\}$	$\{q_2, q_5\}$ IS IS	$\{q_3, q_6\}$ IS IS
$\{q_2, q_7\}$	$\{q_3, q_6\}$ IS IS	$\{q_1, q_4\}$ FS FS

above two condition
 are satisfied so we
 can say that two automata
 are equivalence to each
 other.



B

Equivalence of two finite automata



If initial & final state is same in one automata then the initial & final state is also same in another automata

A → q1 initial & final
B → q4 → initial & final

	c	d
{q1, q4}	{q1, q4} FS FS	{q2, q5} IS IS
{q2, q5}	q3, q7 IS IS	q1, q6 FS IS

→ this prove that A & B are equivalent.

Pigeon hole principle

- If n Pigeon are assigned to m Pigeonhole then atleast one Pigeon hole contain two or more Pigeon $m < n$
- If one pPigeon is aassigned to each Pigeonhole then $n-m$ pigeon are left without assigned to any Pigeon hole

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Thank You

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